

Related U.S. Application Data

continuation of application No. 14/940,323, filed on Nov. 13, 2015, now Pat. No. 10,219,863.

- (60) Provisional application No. 62/079,621, filed on Nov. 14, 2014.

(51) **Int. Cl.**

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A61B 18/00 (2006.01)

A61B 18/20 (2006.01)

A61B 18/22 (2006.01)

(52) **U.S. Cl.**

CPC *A61B 5/0075* (2013.01); *A61B 5/0084* (2013.01); *A61B 2018/00172* (2013.01); *A61B 2018/00642* (2013.01); *A61B 2018/00982* (2013.01); *A61B 2018/2035* (2013.01); *A61B 2018/2065* (2013.01); *A61B 2018/208* (2013.01); *A61B 2018/2211* (2013.01); *A61B 2018/2244* (2013.01)

(56) **References Cited**

U.S. PATENT DOCUMENTS

- 4,718,417 A * 1/1988 Kittrell G02B 6/4296
606/7
- 4,932,747 A 6/1990 Russell et al.
- 5,192,278 A 3/1993 Hayes et al.
- 5,268,978 A 12/1993 Po et al.
- 5,363,463 A 11/1994 Kleinerman
- 5,373,576 A 12/1994 Minns et al.
- 5,394,492 A 2/1995 Hwang et al.
- 5,513,195 A 4/1996 Opower et al.
- 5,668,903 A 9/1997 Neuberger et al.
- 5,708,747 A 1/1998 Danckwerth
- 5,729,568 A * 3/1998 Opower B23K 26/064
385/116
- 5,754,716 A 5/1998 Kim et al.
- 6,011,890 A 1/2000 Neuberger
- 6,110,195 A * 8/2000 Xie A61B 18/203
606/11
- 6,167,075 A * 12/2000 Craig H01S 3/094003
372/75
- 6,201,989 B1 3/2001 Whitehead et al.
- 6,391,022 B1 5/2002 Furumoto et al.
- 6,400,875 B1 6/2002 Lincoln et al.
- 6,519,387 B1 2/2003 Sunagawa et al.
- 6,882,429 B1 4/2005 Weitekamp et al.
- 7,329,887 B2 * 2/2008 Henson F21V 29/70
359/726
- 7,456,805 B2 11/2008 Ouderkirk et al.
- 7,502,537 B2 3/2009 Kurahashi
- 8,047,985 B2 11/2011 Harris
- 8,158,957 B2 4/2012 Nelson et al.
- 8,320,724 B2 11/2012 Sasaoka
- 8,588,267 B1 11/2013 Panak
- 9,161,684 B2 * 10/2015 Seibel A61B 1/015
- 9,599,507 B2 3/2017 Pawluczyk et al.
- 2001/0055462 A1 12/2001 Seibel
- 2002/0097400 A1 7/2002 Jung et al.
- 2002/0138072 A1 * 9/2002 Black A61B 18/22
606/17
- 2003/0109787 A1 * 6/2003 Black A61B 5/0071
600/473
- 2003/0109860 A1 * 6/2003 Black A61B 18/20
606/17
- 2003/0181893 A1 * 9/2003 Neuberger A61B 18/203
606/9

- 2003/0214571 A1 11/2003 Ishikawa et al.
- 2004/0039242 A1 2/2004 Tolkoff et al.
- 2004/0082940 A1 4/2004 Black et al.
- 2004/0150828 A1 8/2004 Hendrix et al.
- 2004/0218635 A1 11/2004 Schlueter et al.
- 2004/0240489 A1 12/2004 Teramura et al.
- 2005/0152655 A1 7/2005 Sunagawa
- 2005/0279354 A1 12/2005 Deutsch et al.
- 2007/0038124 A1 * 2/2007 Fulghum, Jr. A61B 5/0075
600/476
- 2007/0041083 A1 2/2007 Di Teodoro et al.
- 2007/0060984 A1 3/2007 Webb et al.
- 2007/0154153 A1 * 7/2007 Fomitchev G02B 6/06
385/115
- 2007/0219601 A1 * 9/2007 Neuberger A61B 18/24
607/89
- 2007/0291265 A1 * 12/2007 Holman G01N 21/532
702/22
- 2008/0310181 A1 12/2008 Gurevich et al.
- 2009/0109698 A1 4/2009 Koyata et al.
- 2009/0275927 A1 * 11/2009 Fein H01S 3/2232
606/3
- 2010/0228089 A1 * 9/2010 Hoffman H01S 5/06808
372/38.07
- 2010/0278503 A1 11/2010 Nakai et al.
- 2011/0134519 A1 6/2011 Cooper
- 2011/0135260 A1 6/2011 Nakai et al.
- 2011/0196356 A1 * 8/2011 Neuberger A61B 18/18
606/15
- 2011/0295347 A1 * 12/2011 Wells A61N 5/0622
607/89
- 2011/0306956 A1 * 12/2011 Islam A61B 18/24
606/14
- 2012/0010465 A1 * 1/2012 Erikawa A61B 1/0653
600/109
- 2012/0101374 A1 4/2012 Tearney et al.
- 2012/0123399 A1 * 5/2012 Belikov A61B 18/22
606/17
- 2012/0232534 A1 9/2012 Rink et al.
- 2012/0307512 A1 12/2012 Cogger et al.
- 2013/0006119 A1 * 1/2013 Pan A61N 5/0616
600/476
- 2013/0336343 A1 12/2013 Miyabe et al.
- 2014/0071521 A1 3/2014 Liu
- 2014/0078378 A1 3/2014 Demers et al.
- 2015/0062573 A1 3/2015 Liu et al.
- 2015/0216398 A1 8/2015 Yang et al.
- 2015/0289936 A1 * 10/2015 Sotak A61B 18/245
606/7
- 2016/0101263 A1 4/2016 Blumenkranz et al.
- 2016/0357007 A1 12/2016 Swanson
- 2016/0363728 A1 12/2016 Wang et al.
- 2017/0071509 A1 3/2017 Pandey et al.
- 2017/0082805 A1 3/2017 Saikawa et al.
- 2017/0224220 A1 8/2017 Tunnell et al.
- 2018/0157039 A1 6/2018 Yu et al.
- 2018/0281108 A1 10/2018 Victor et al.

FOREIGN PATENT DOCUMENTS

- EP 2241280 A2 10/2010
- JP H09-173346 A 12/1995
- JP 2005-513762 5/2005
- WO WO 2014/074678 A1 5/2014

OTHER PUBLICATIONS

International Search Report and Written Opinion for corresponding International Application No. PCT/US2015/060547, dated Mar. 3, 2016 (9 pages).

* cited by examiner

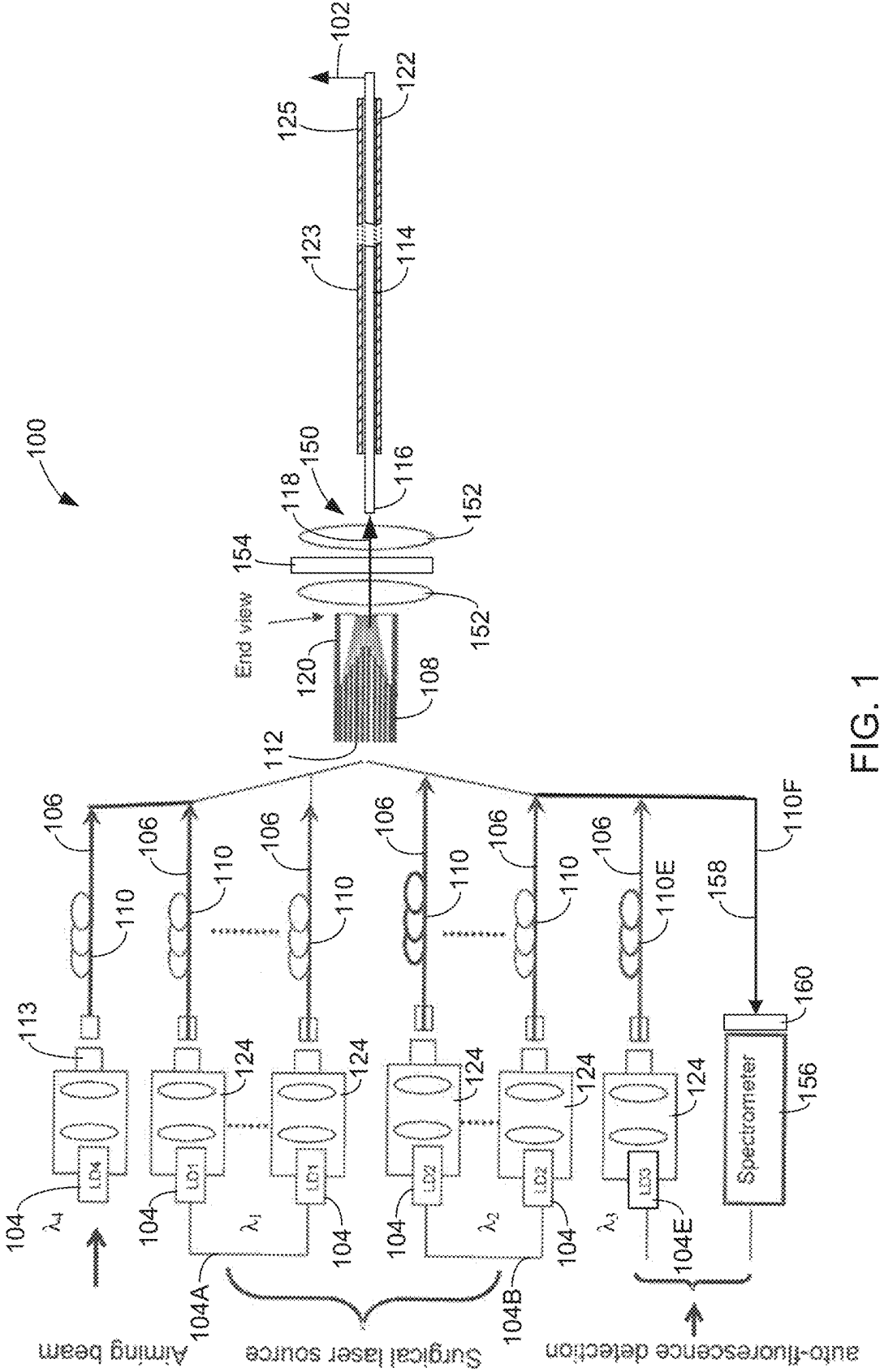


FIG. 1

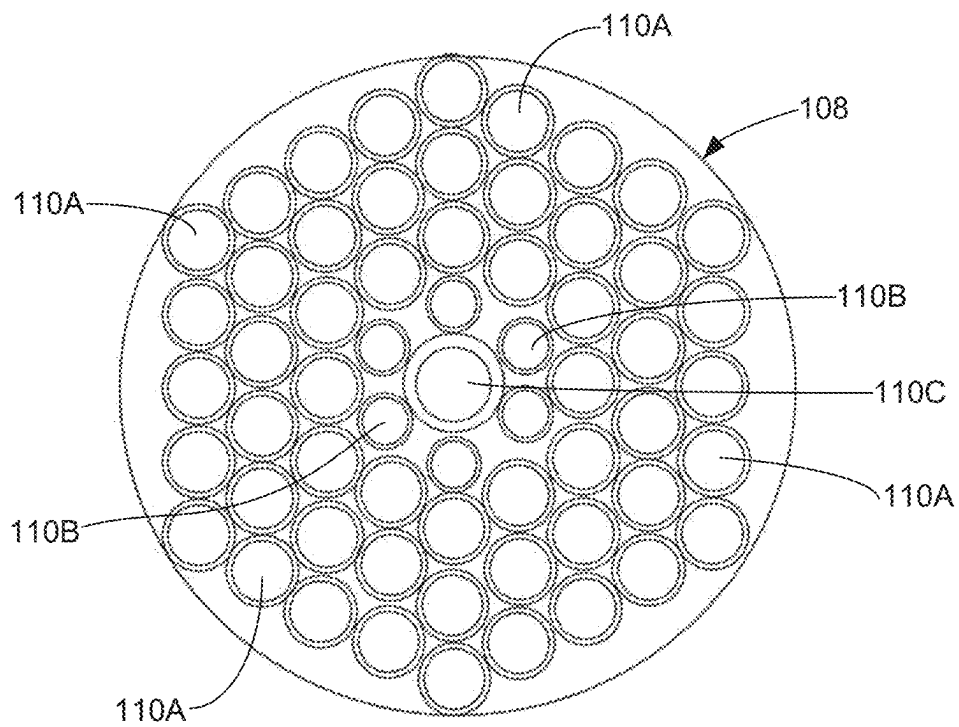


FIG. 2

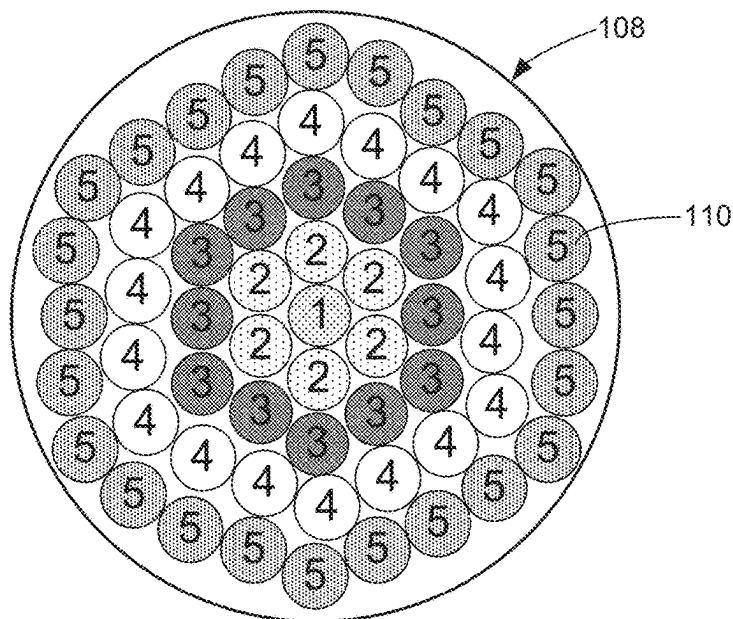


FIG. 3

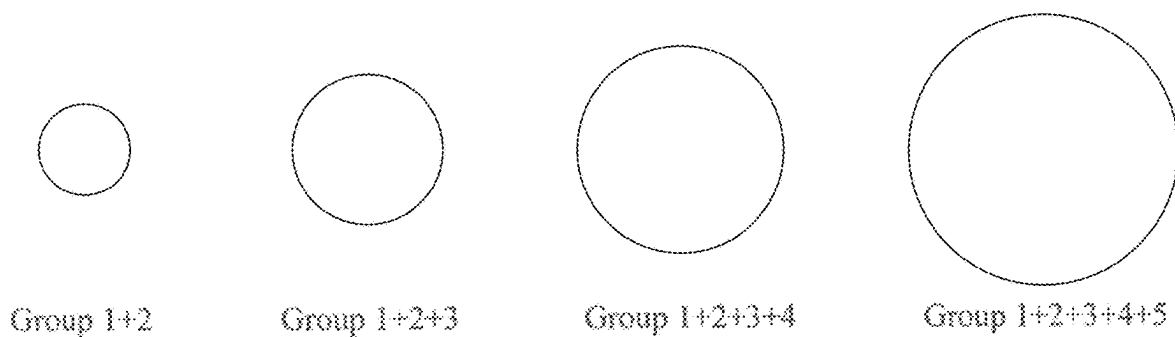


FIG. 4

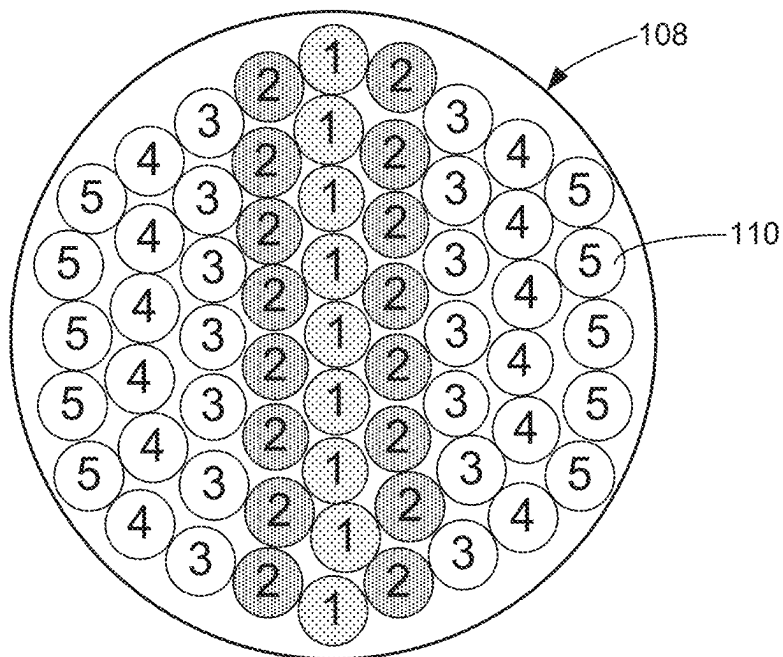


FIG. 5

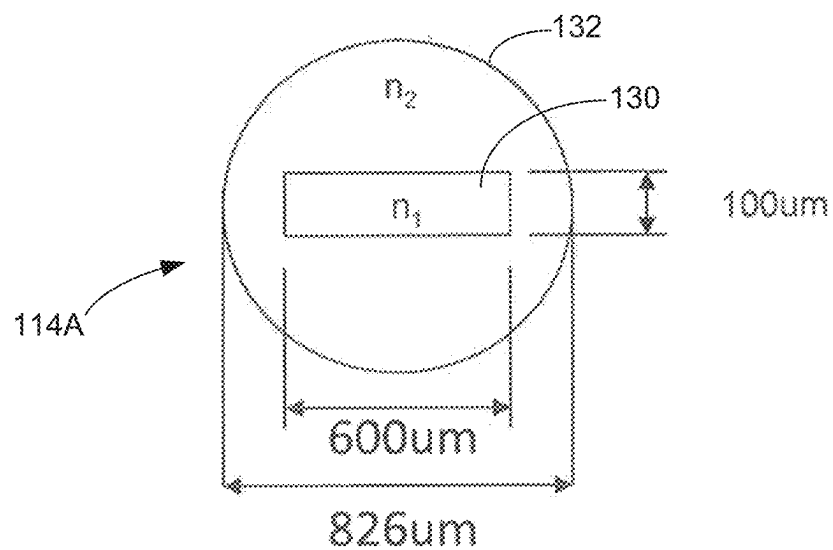


FIG. 6

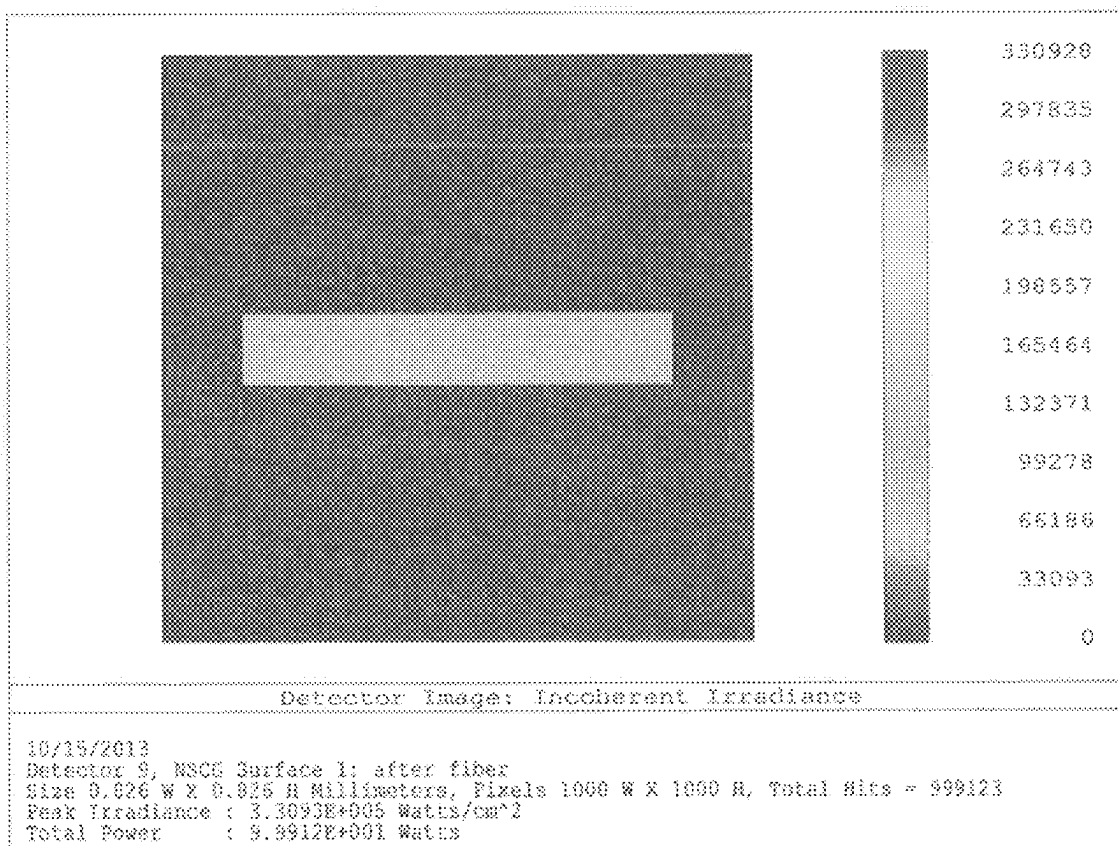


FIG. 7

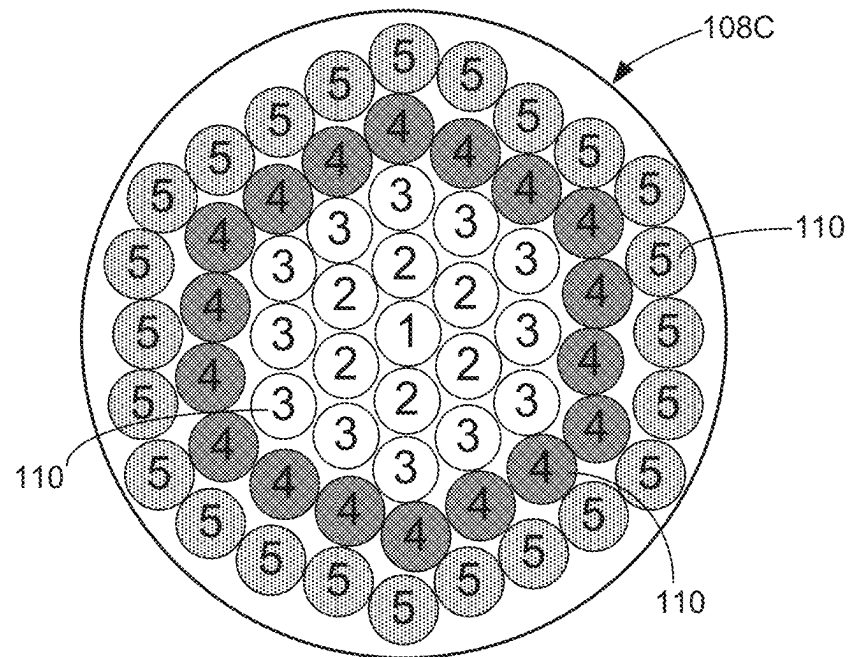


FIG. 8

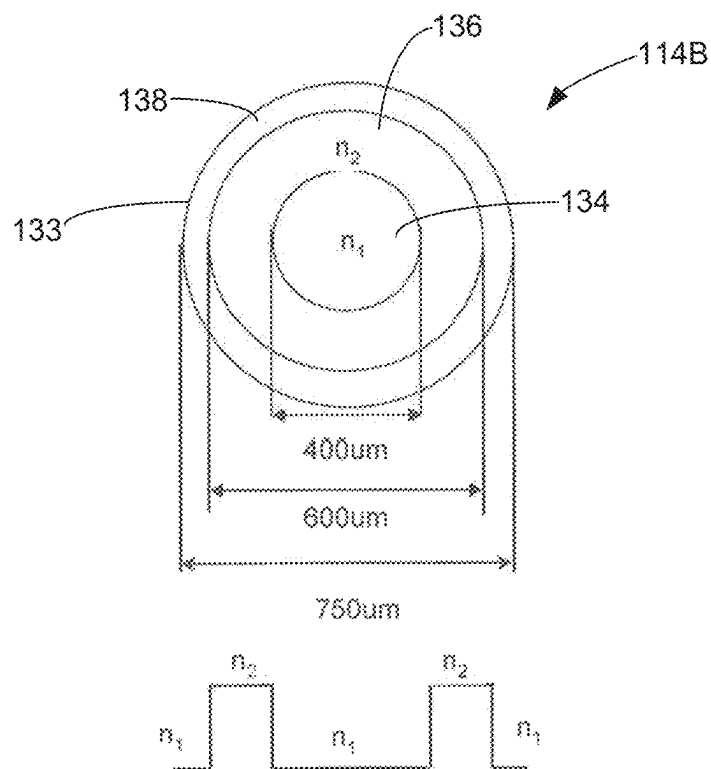


FIG. 9

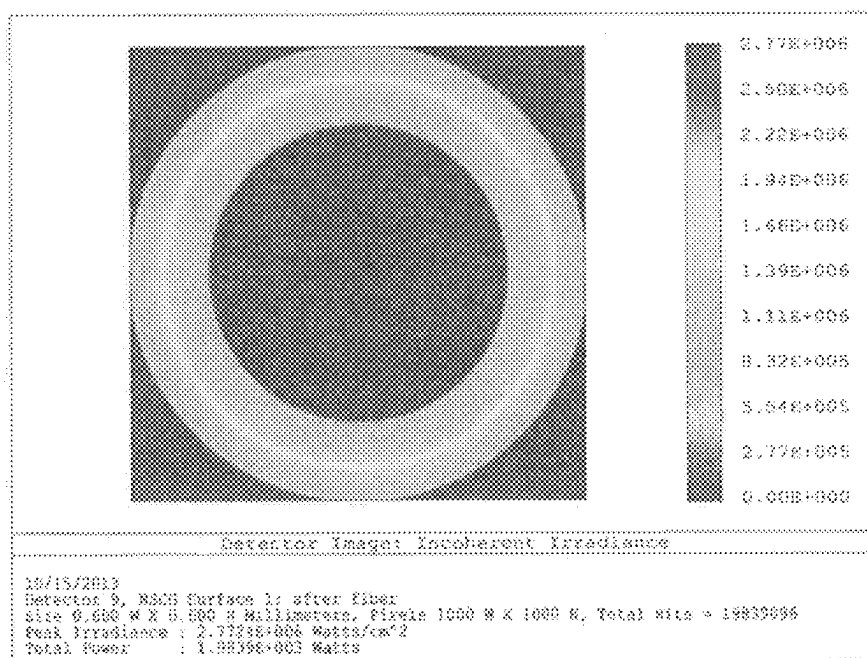


FIG. 10

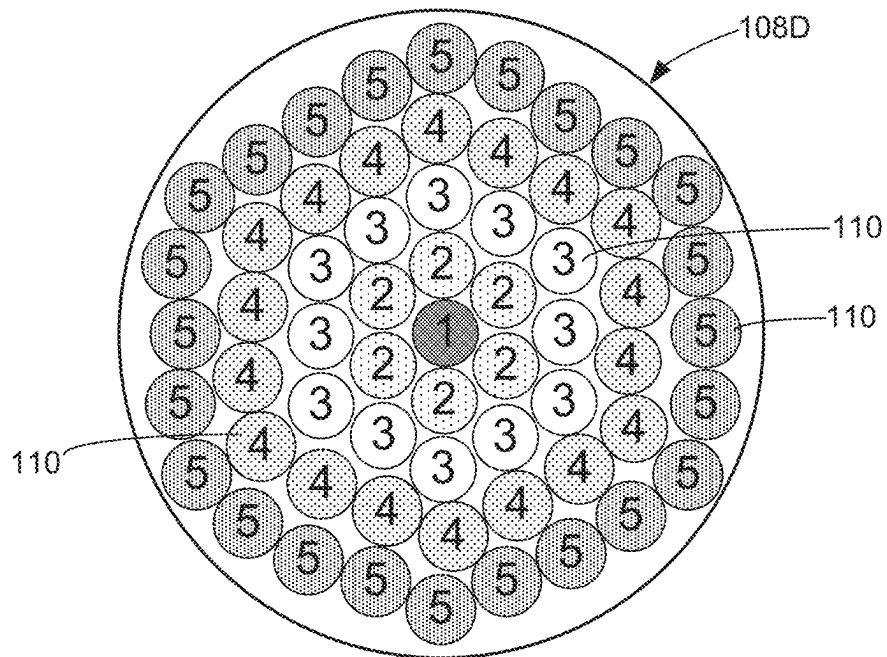


FIG. 11

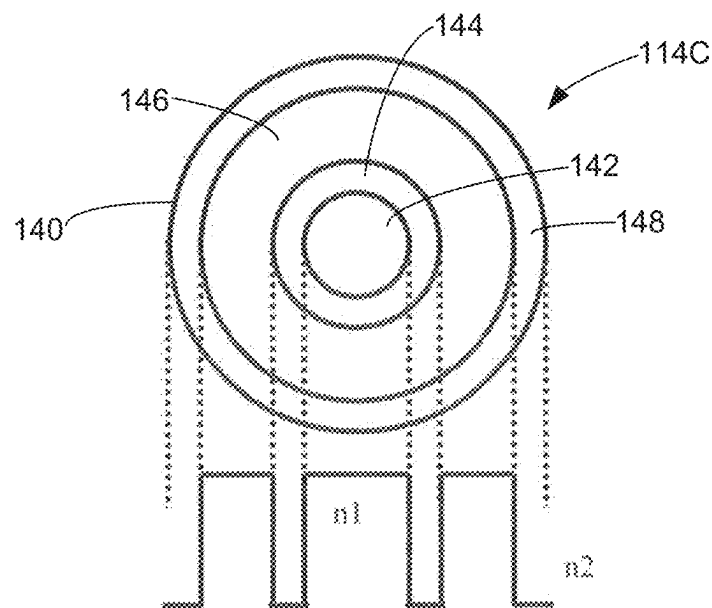


FIG. 12

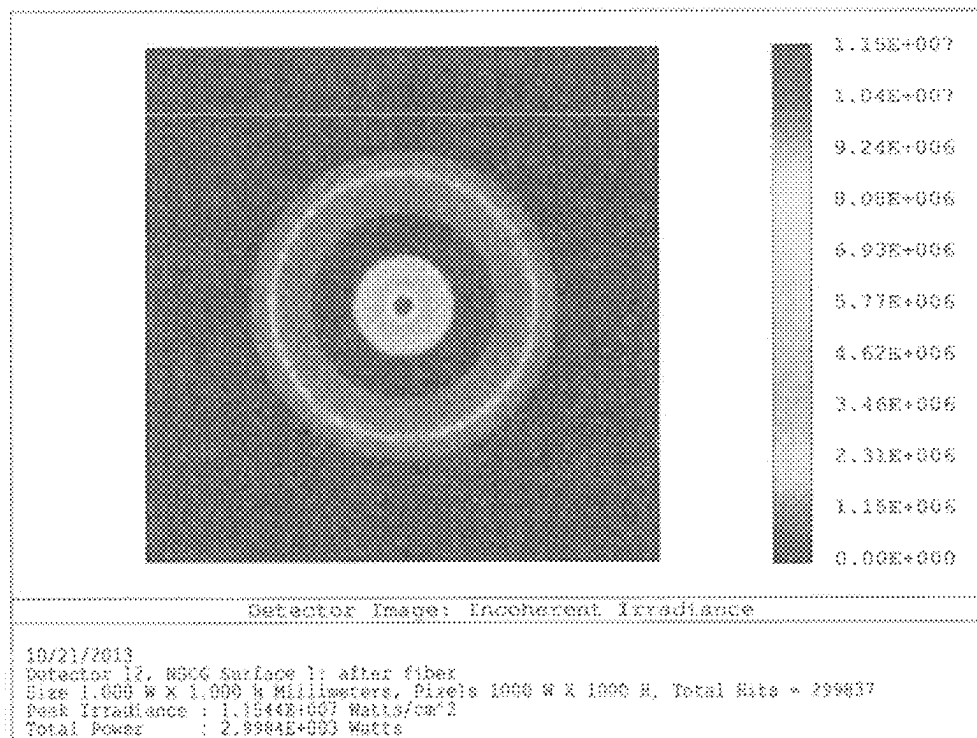


FIG. 13

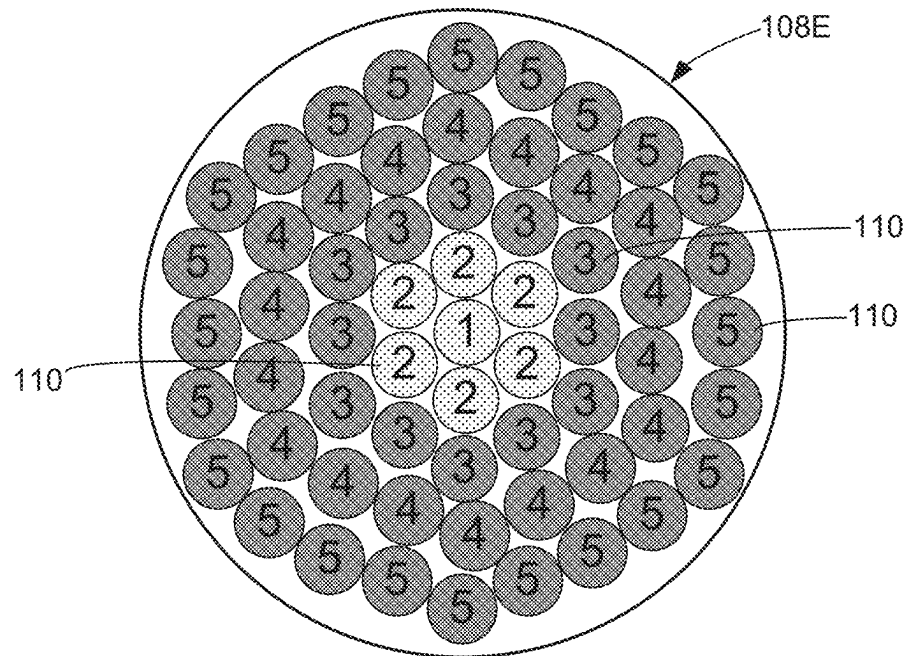


FIG. 14

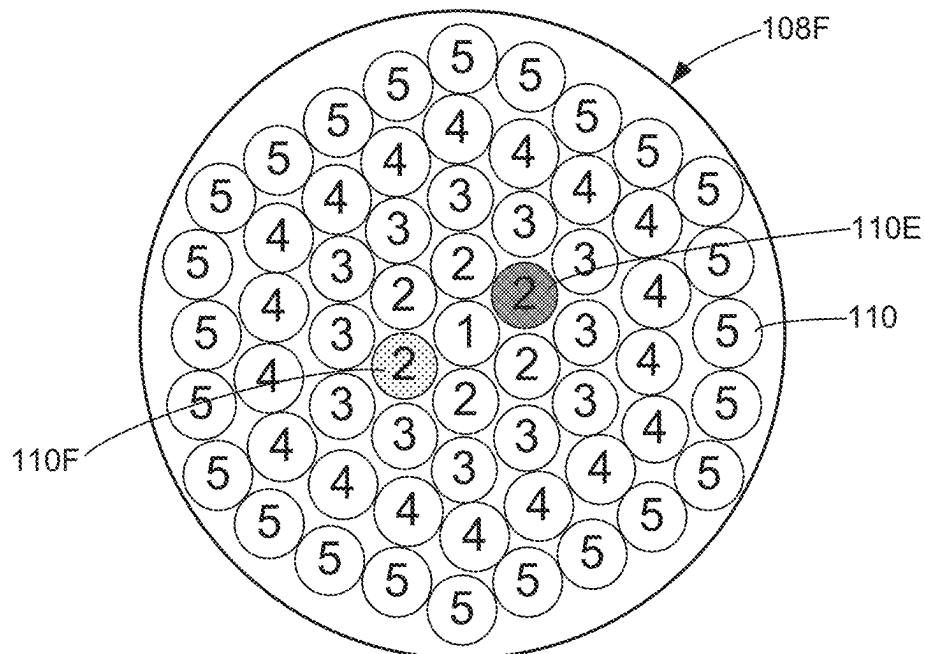
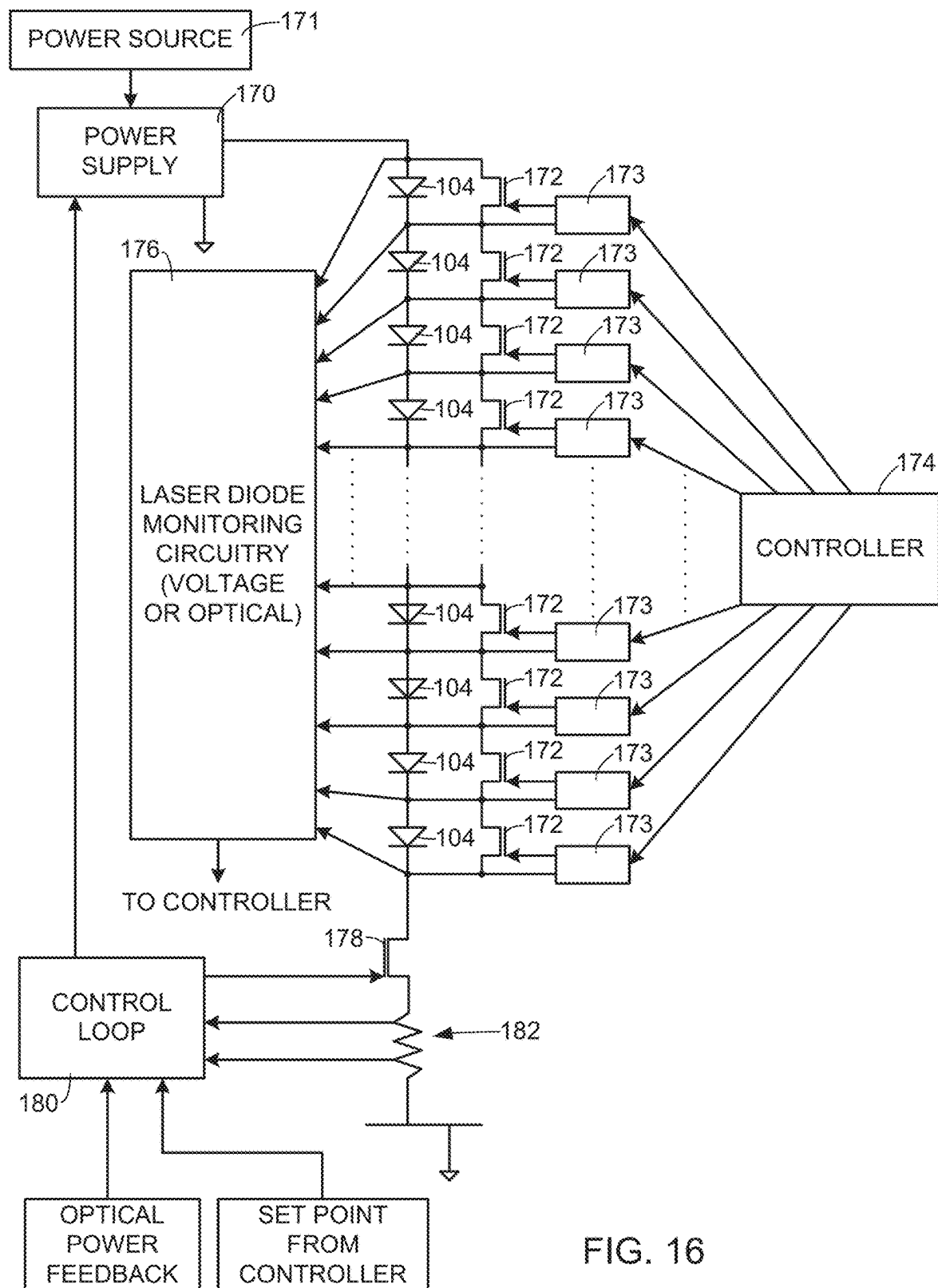


FIG. 15



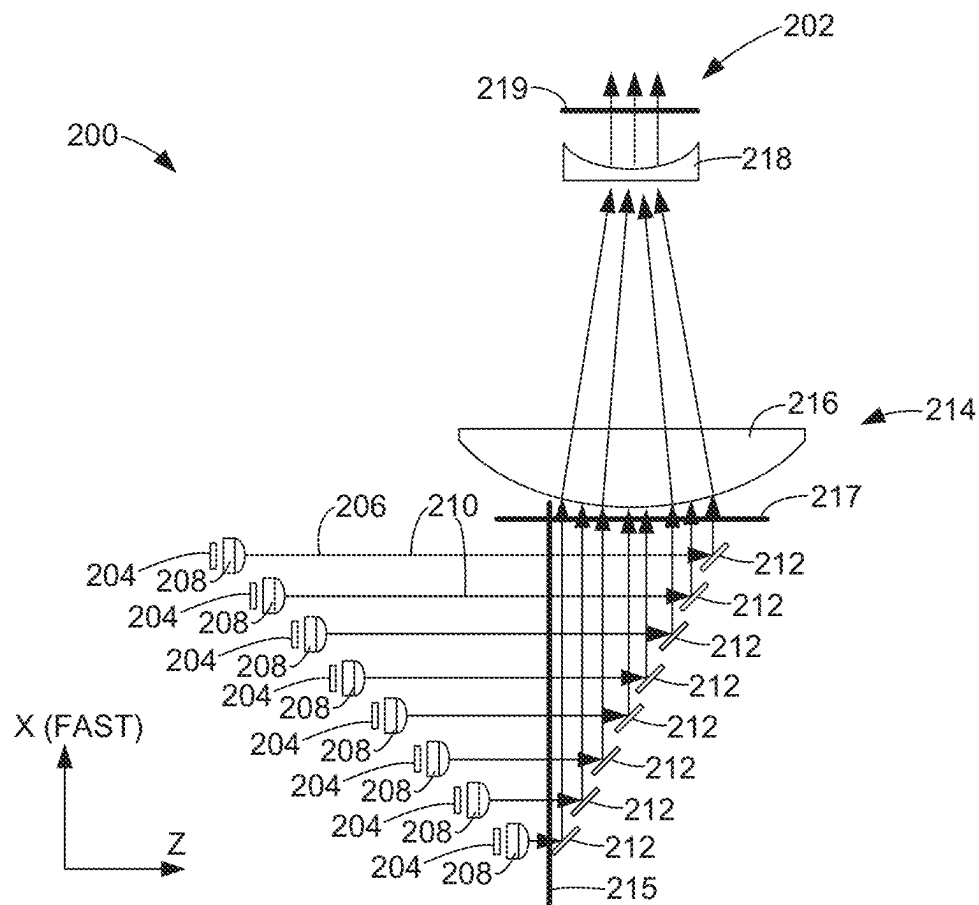


FIG. 17

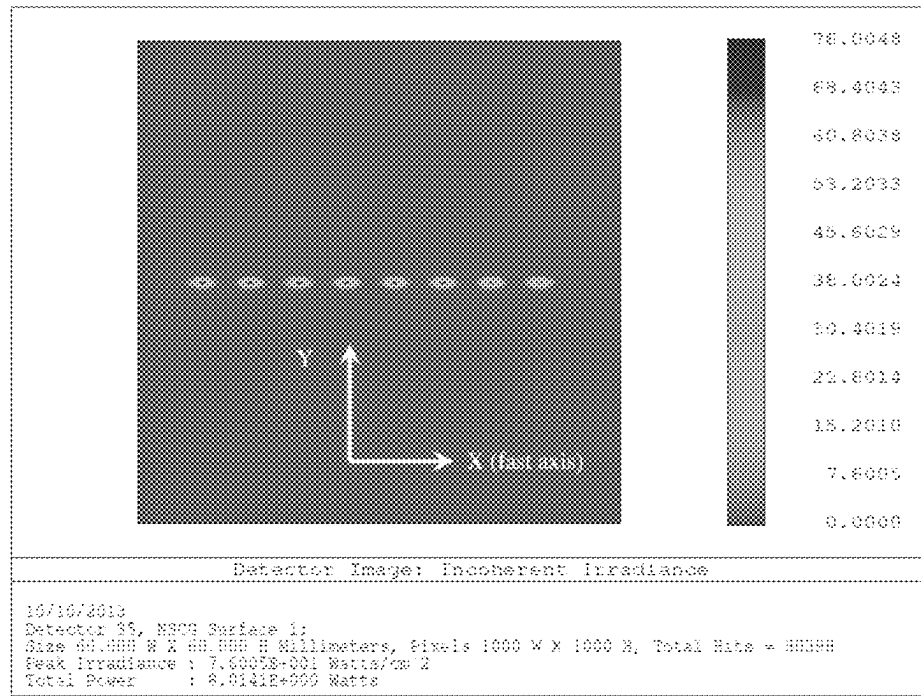


FIG. 18

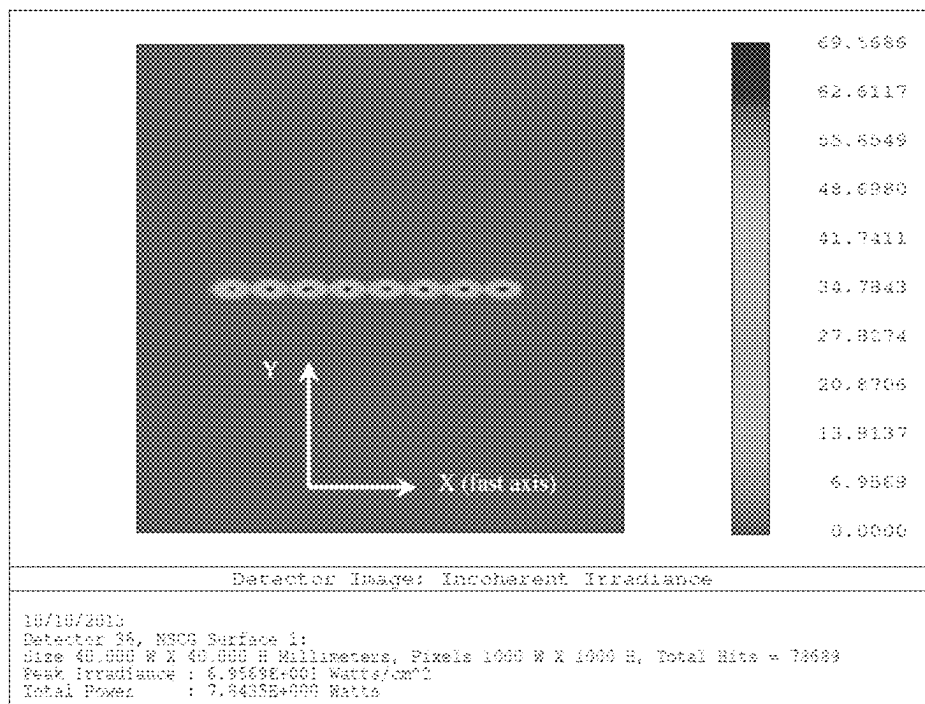


FIG. 19

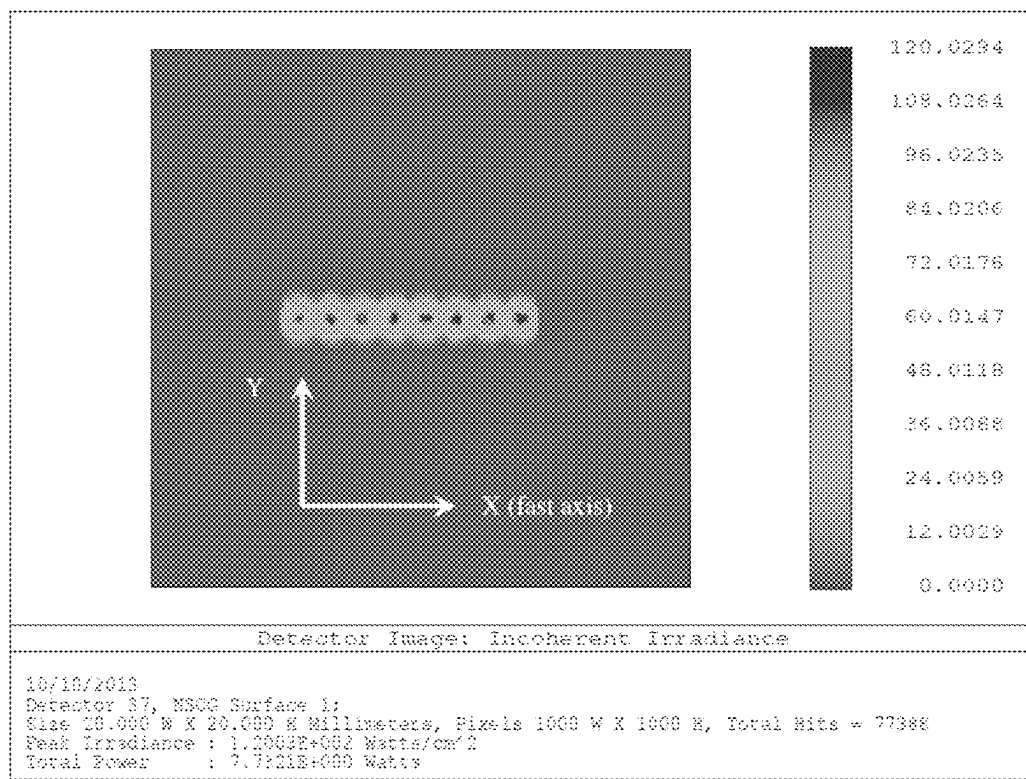


FIG. 20

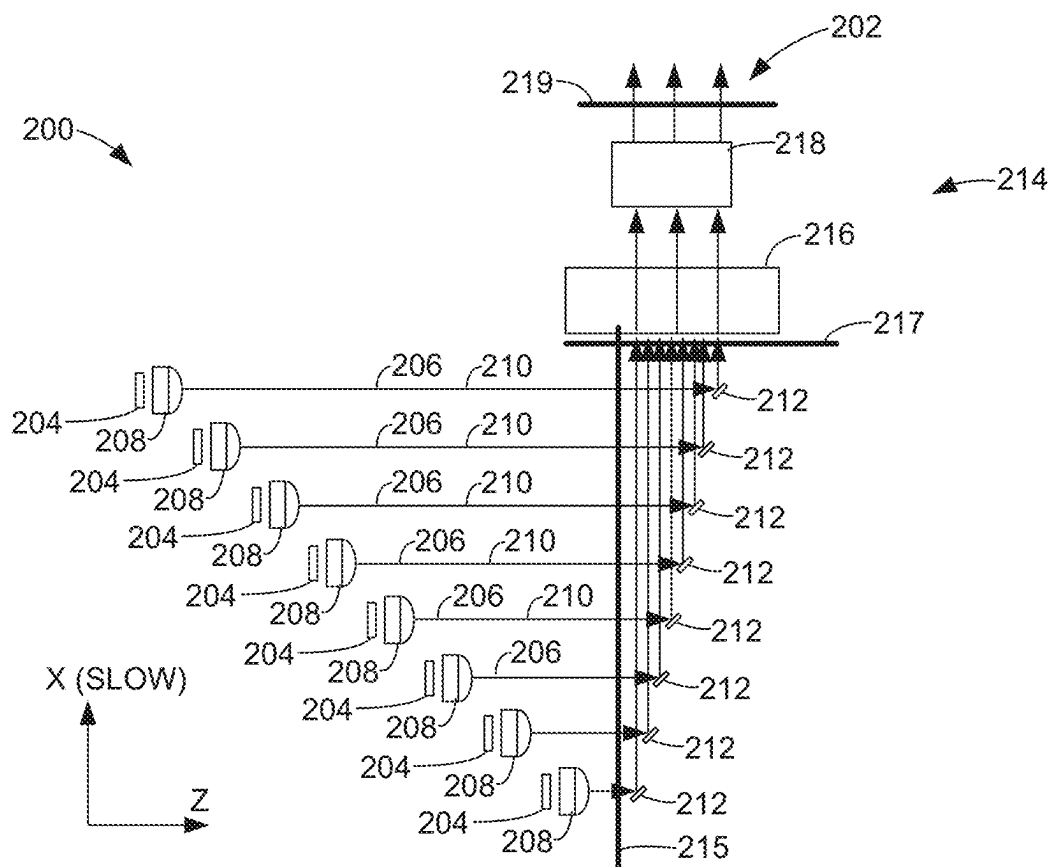


FIG. 21

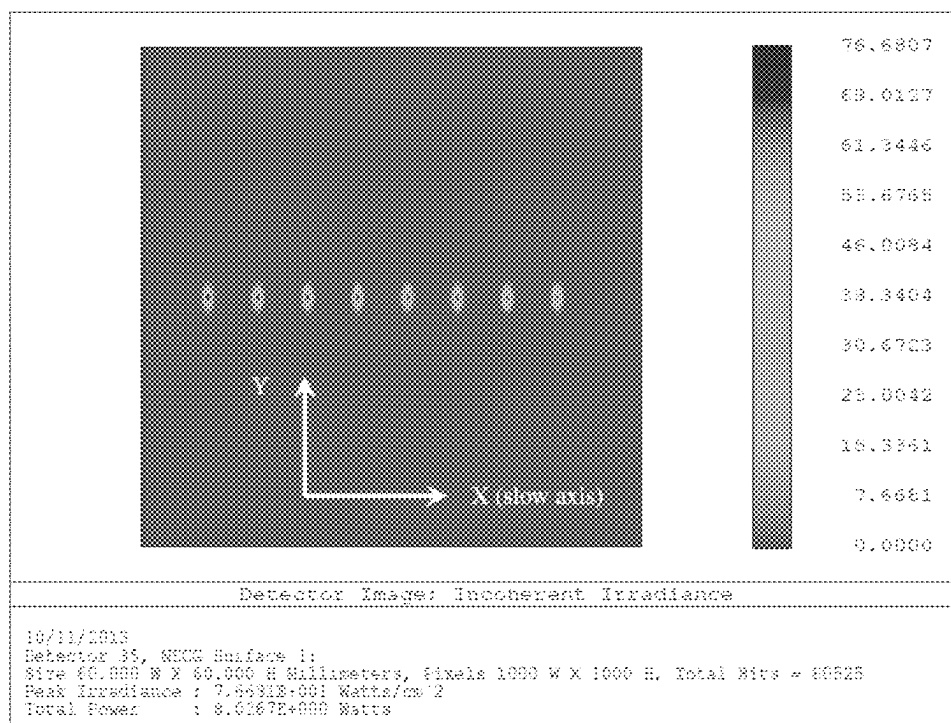


FIG. 22

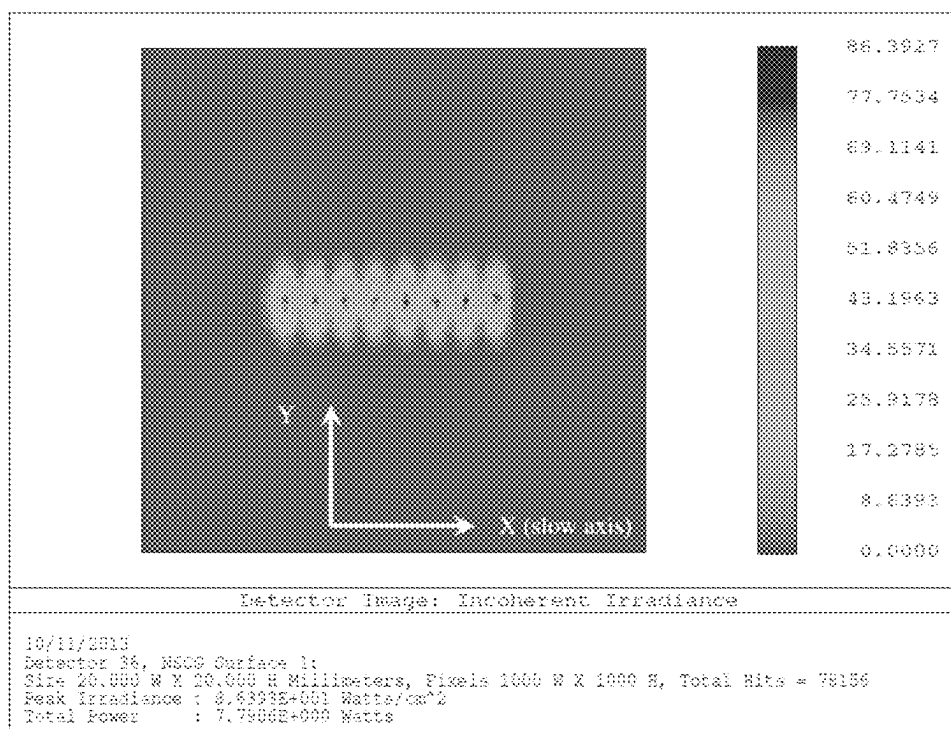


FIG. 23

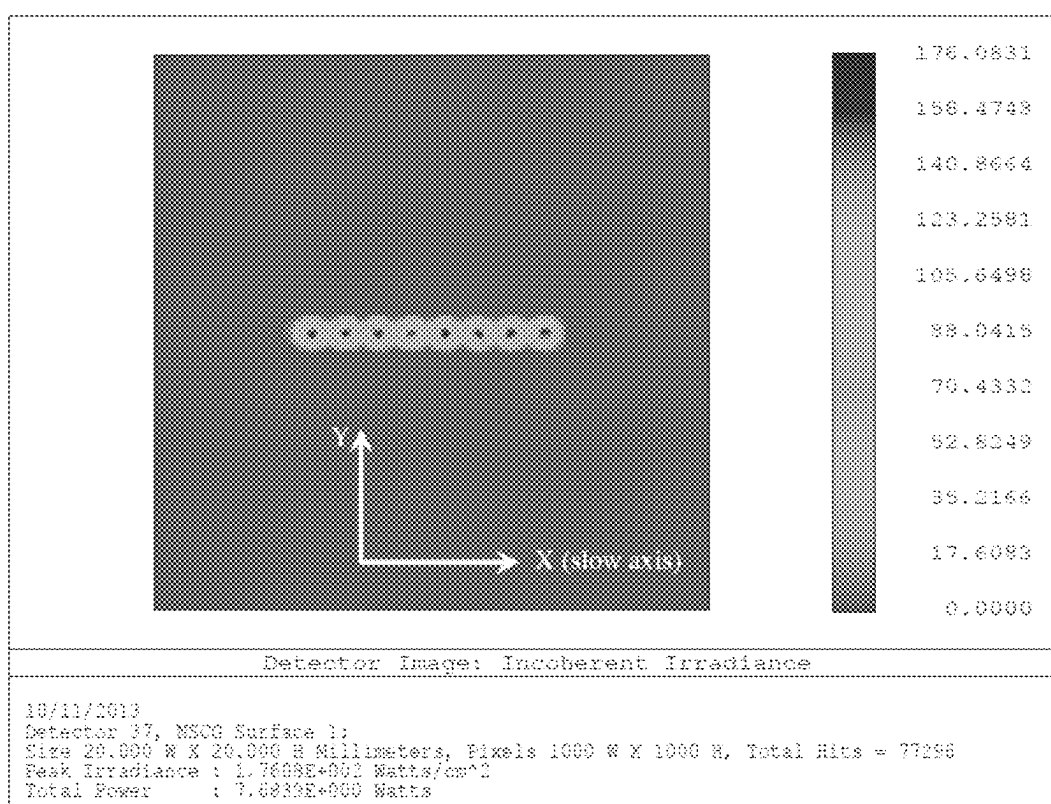


FIG. 24

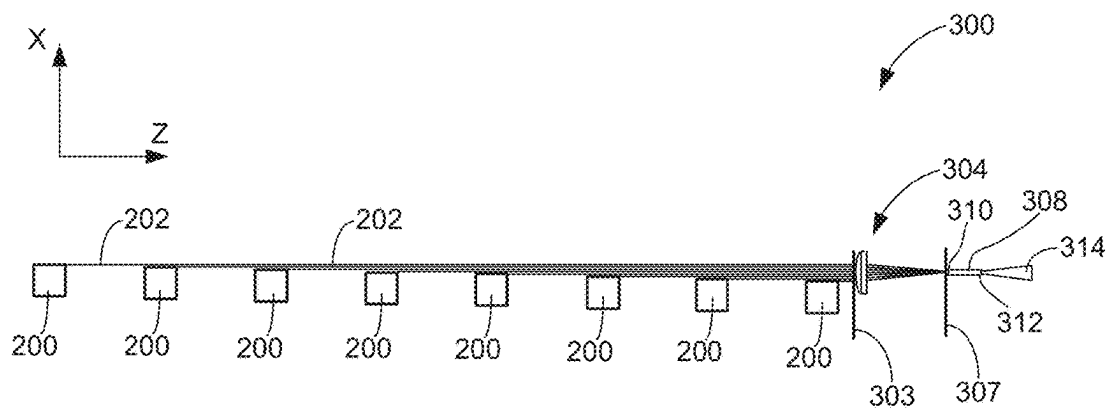


FIG. 25

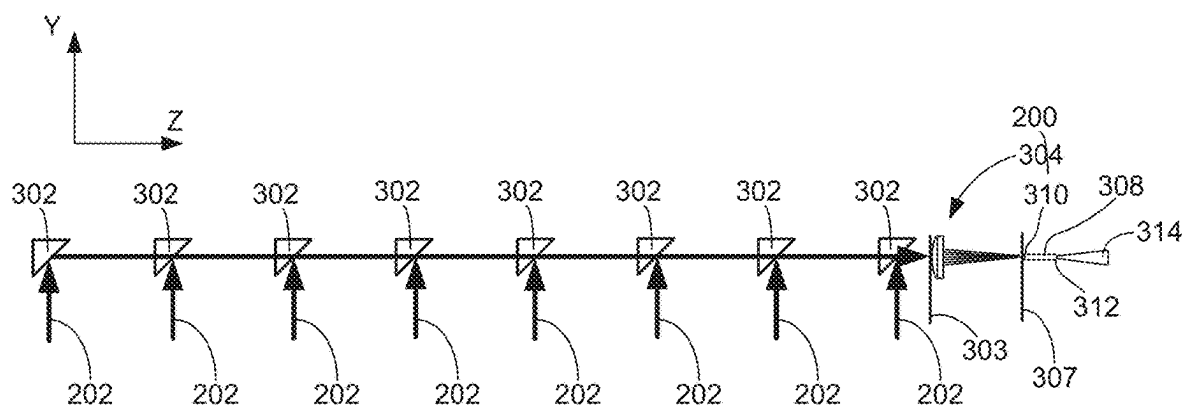


FIG. 26

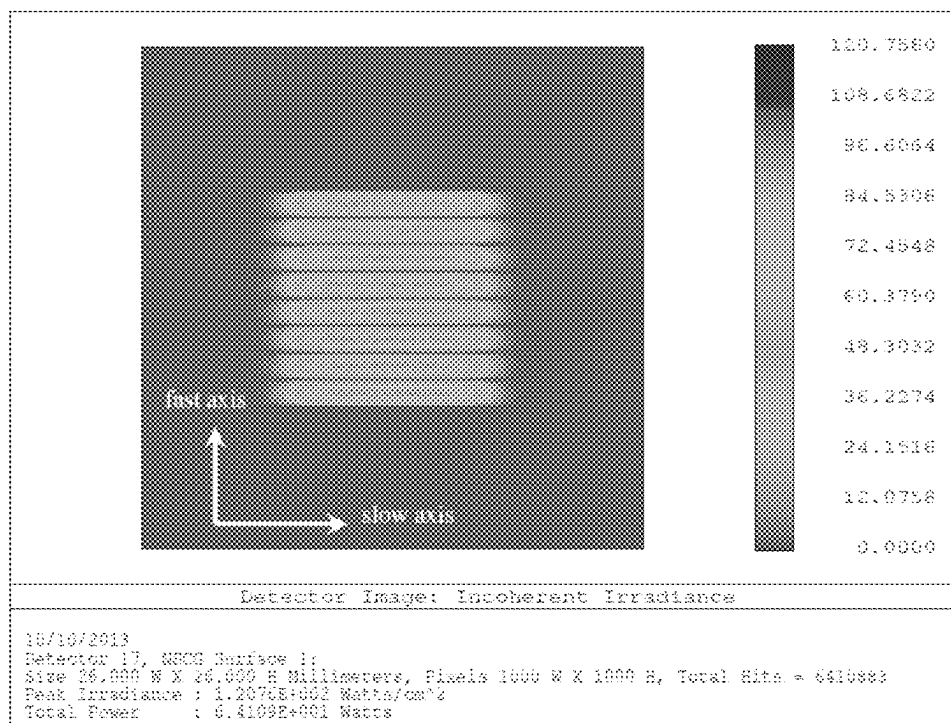


FIG. 27

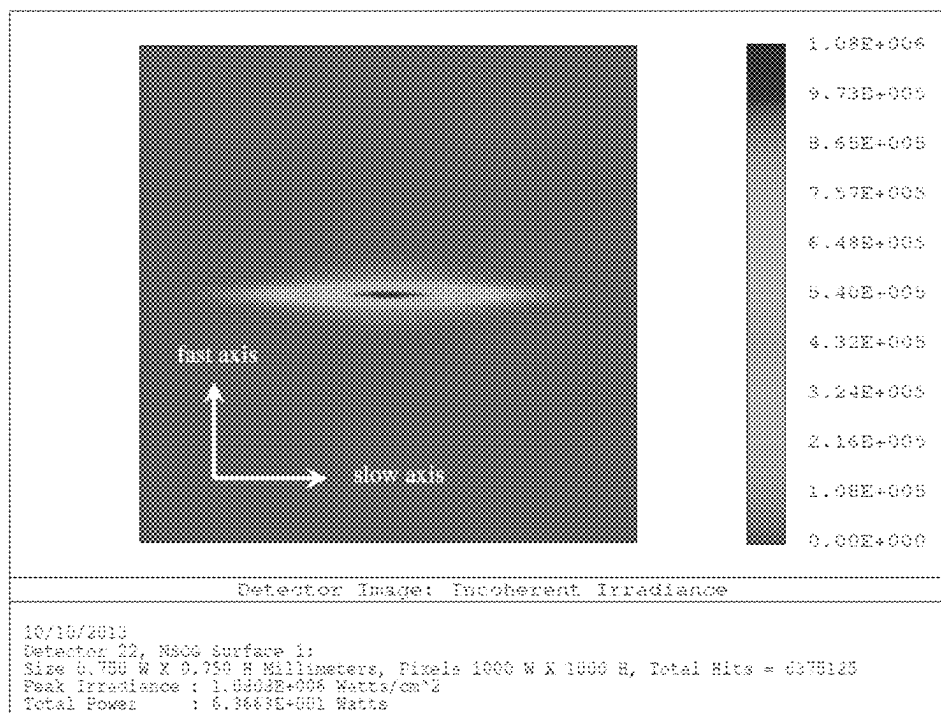


FIG. 28

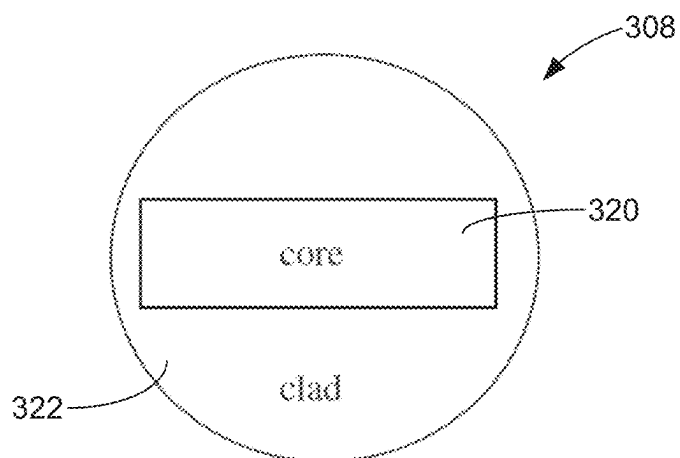


FIG. 29

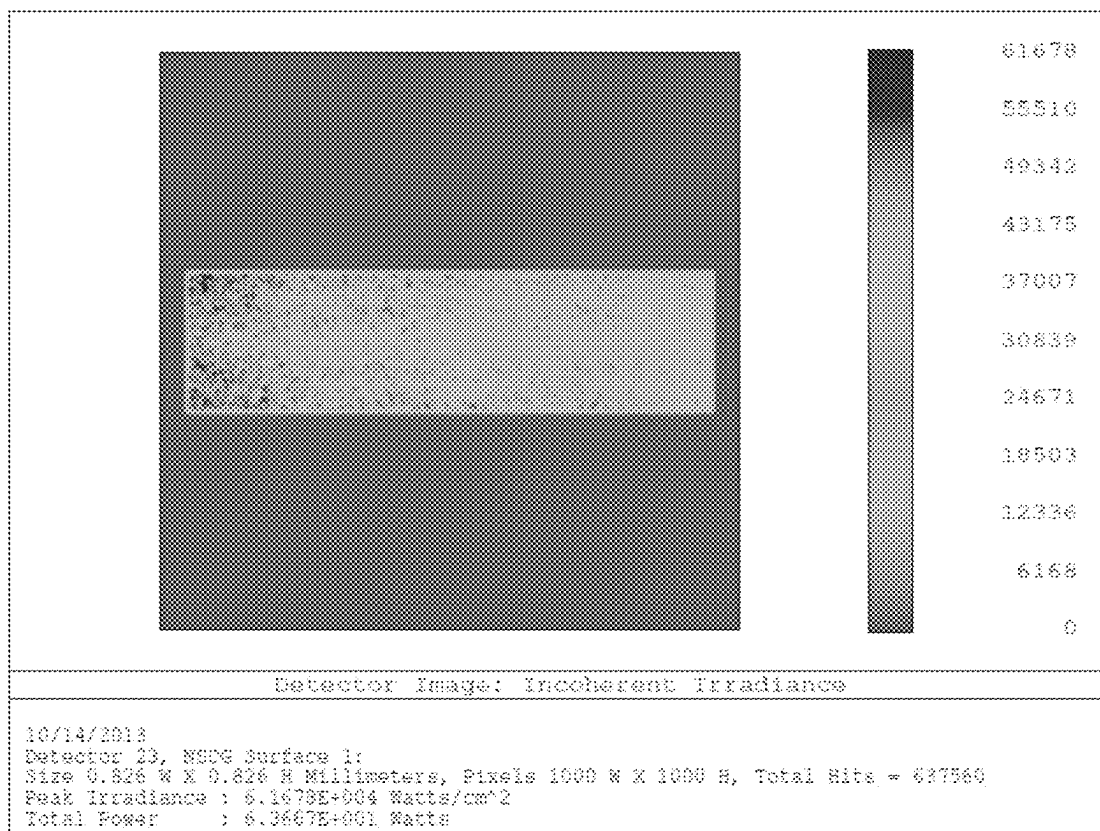


FIG. 30

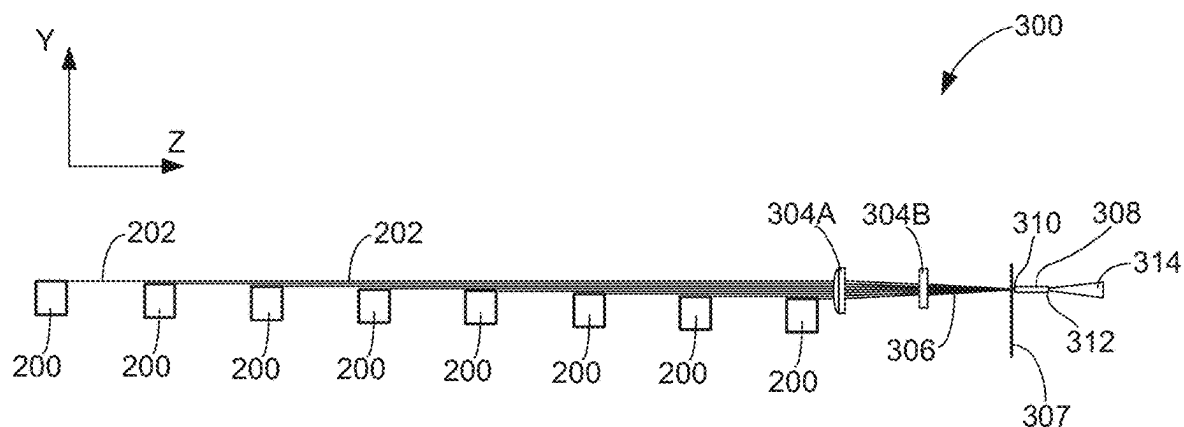


FIG. 31

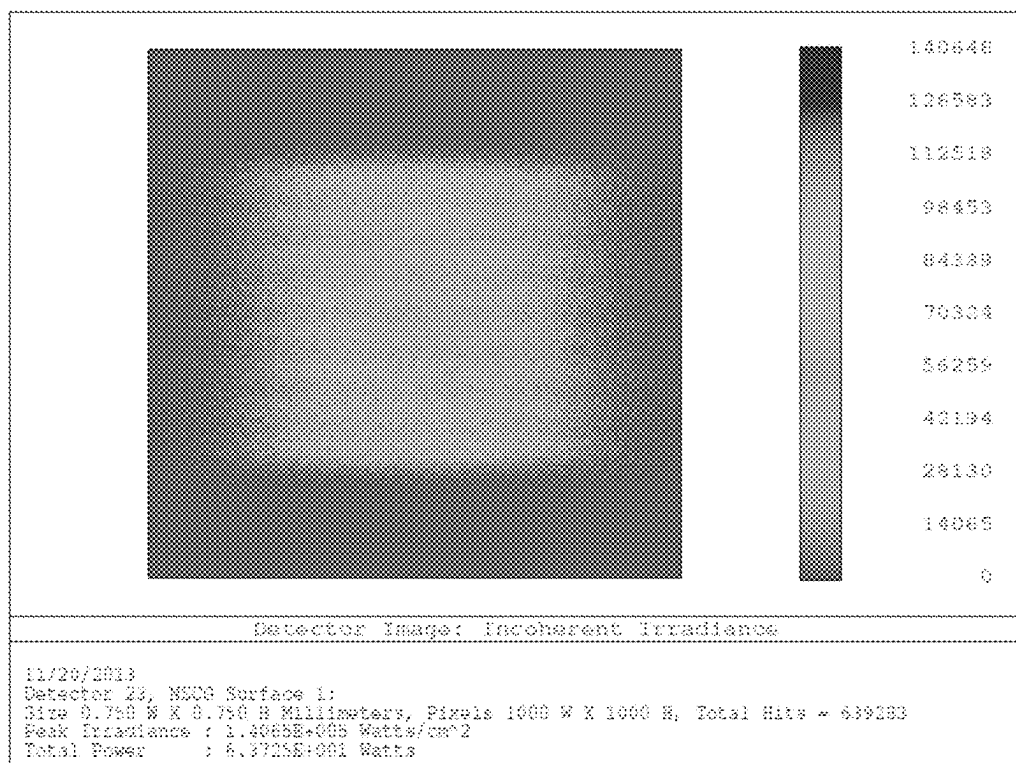


FIG. 32

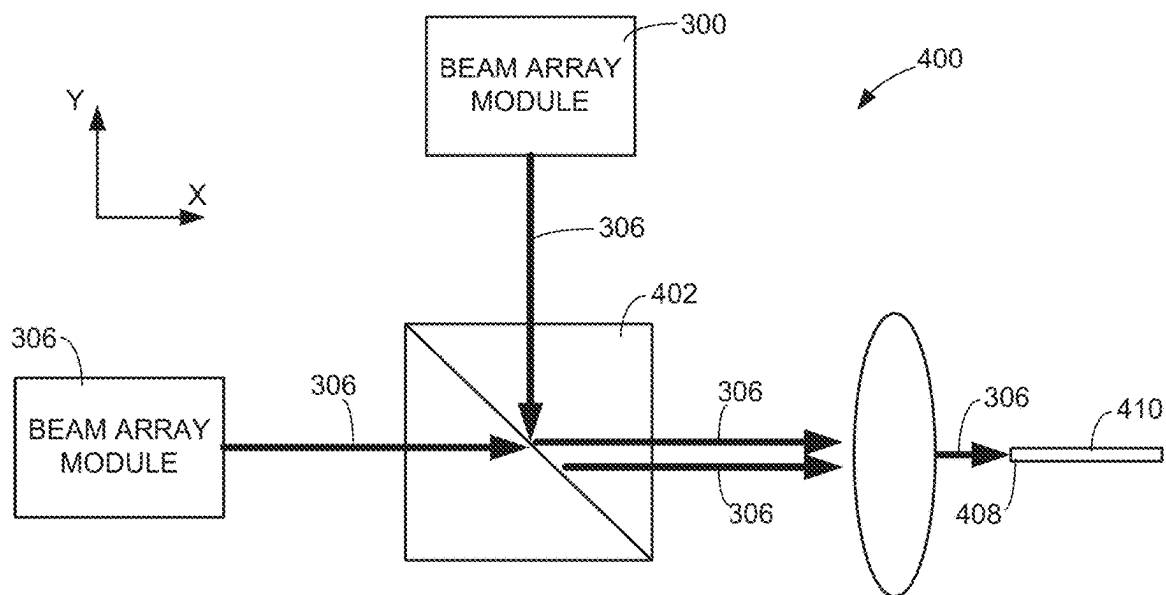


FIG. 33

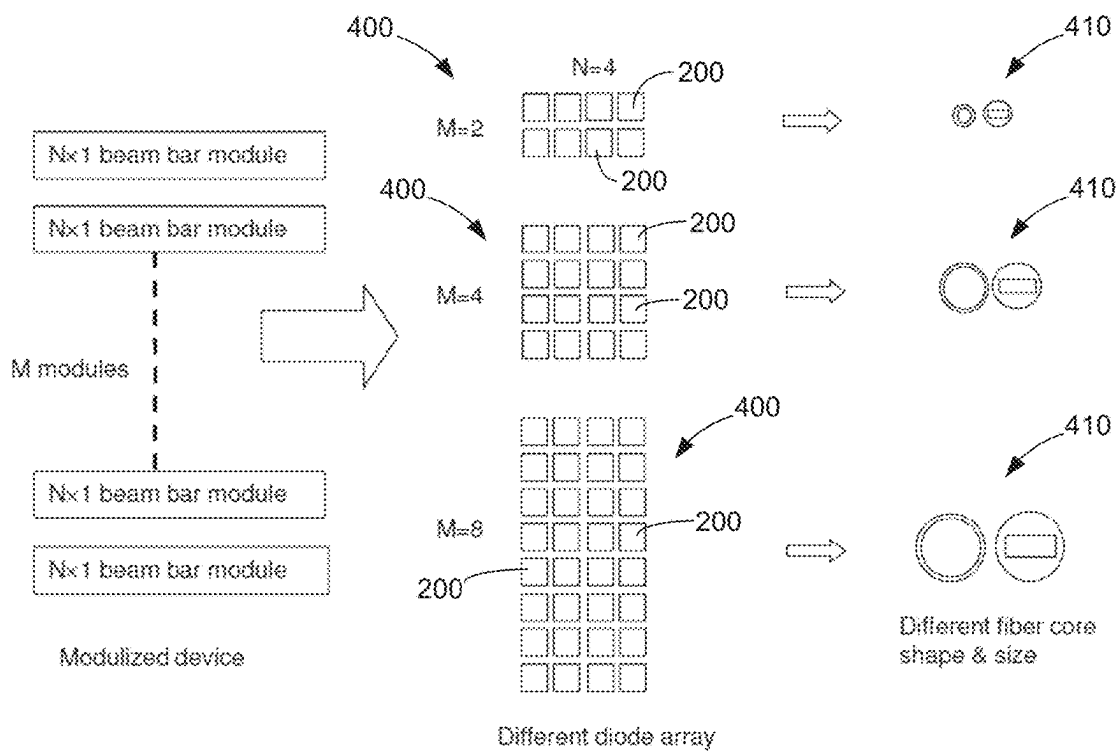


FIG. 34

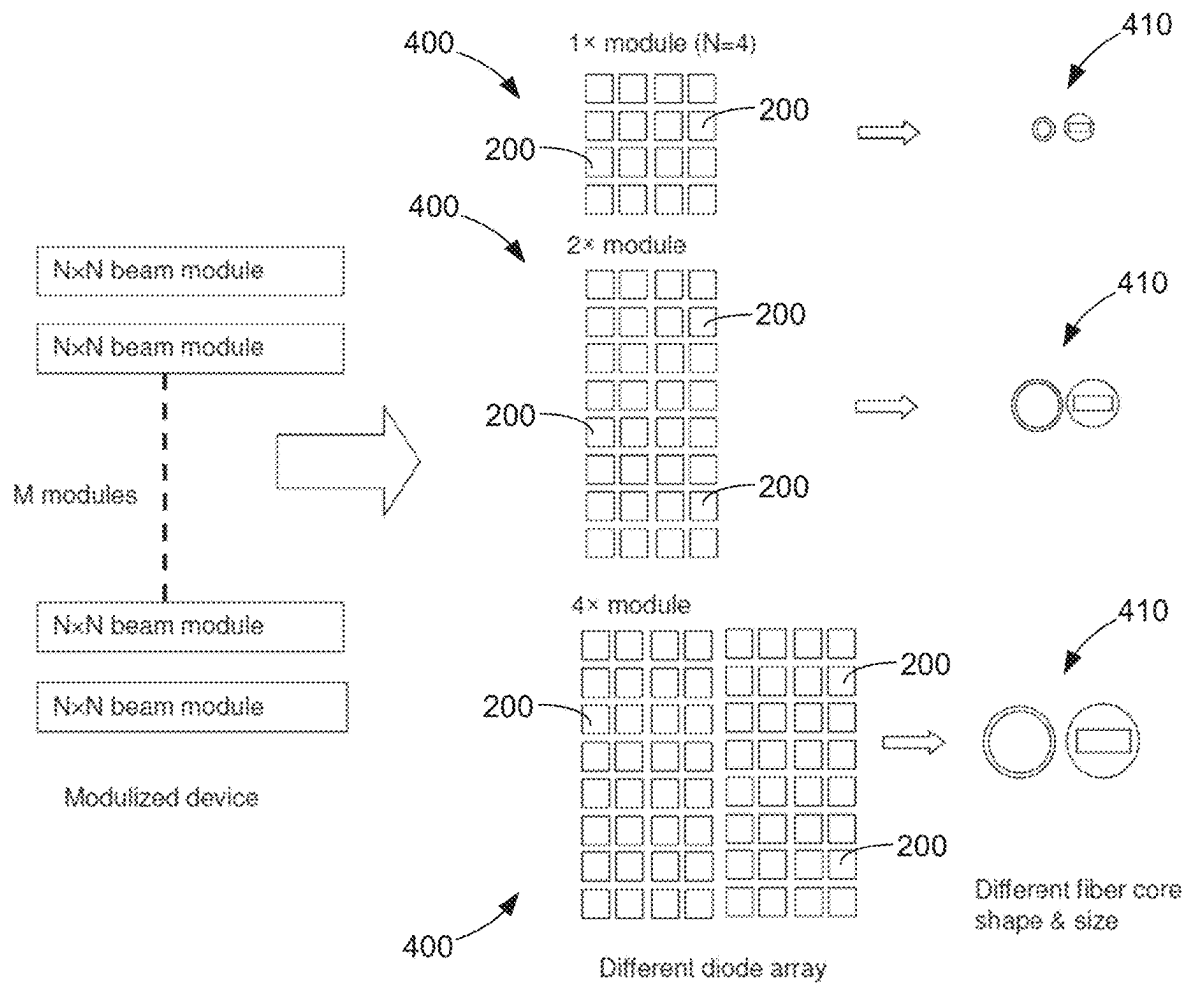


FIG. 35

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SURGICAL LASER SYSTEMS AND LASER DEVICES

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. application Ser. No. 16/251,278, filed Jan. 18, 2019, now U.S. Pat. No. 11,213,351, which is a continuation of U.S. application Ser. No. 14/940,323, filed Nov. 13, 2015, now U.S. Pat. No. 10,219,863, which is based on and claims the benefit of priority of U.S. Provisional Patent Application No. 62/079,621, filed Nov. 14, 2014, the content of each of which are hereby incorporated by reference in its entirety.

BACKGROUND

Embodiments of the invention generally relate to laser devices including, for example, laser systems, laser bars and laser modules comprising laser diodes, and methods of using the laser devices.

Lasers have been increasingly adopted as medical surgical tools and optical fibers have been normally used as delivery devices. As compared to traditional surgical tools, laser surgery can reduce bleeding, pain and infection. Additionally, patients often have less hospitalization time after laser surgery.

High power and high brightness fiber-coupled diode lasers have been increasingly adopted in industrial and medical applications because of their intrinsically simple design, low cost and high wall plug efficiency. Laser diode bars, which comprise multiple laser diodes, have been the common building blocks for the high power laser systems. However, for some wavelength ranges, laser diode bars are not available. Thus, it is necessary to utilize only single semiconductor laser diode emitters or semiconductor lasers (hereinafter "laser diodes") for these wavelength ranges.

Due to their low power, it is necessary to combine the output laser energy from multiple laser diodes into an optical fiber to provide the desired power level. However, it can be difficult to combine the laser energy from individual laser diodes into a single composite beam, particularly when it is desired to have a high power composite laser energy beam (e.g., more than 100 W) using low power (e.g., 1-3 W) laser diodes.

Different surgical applications often utilize laser energy having different properties. For example, different surgical applications may require laser energy having different wavelengths, different pulse widths and pulse repetition rates, different beam sizes and shapes, different power intensities and different feedback systems.

Embodiments of the invention provide solutions to these and other problems.

SUMMARY

Embodiments are directed to surgical laser systems and laser devices utilizing a plurality of laser diodes. One embodiment of a surgical laser system includes an array of laser diodes that are configured to output laser energy, a fiber bundle, a delivery fiber, and a tubular sheath. The fiber bundle includes a plurality of optical fibers and has a proximal end that is configured to receive laser energy from the array of laser diodes. The delivery fiber includes a proximal end that is configured to receive laser energy from a distal end of the fiber bundle. The tubular sheath defines a lumen, in which at least a portion of the delivery fiber is

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disposed. The tubular sheath is insertable into a working channel of an endoscope or a cystoscope. A distal end of the tubular sheath is configured to deliver laser energy discharged from the delivery fiber into a body of a patient.

Some embodiments are directed to a method of treating a patient using the above-described surgical laser system. In one embodiment of the method, the tubular sheath is inserted into a body of the patient. A first sub-array of the laser diodes are operated to deliver a first beam of laser energy to a tissue of the patient. The first and a second sub-array of the laser diodes are simultaneously operated to deliver a second beam of laser energy to the tissue of the patient having a different size or shape than the first beam.

Another embodiment is directed to a method of producing a laser beam using a surgical laser system. In the method, a discrete beam of laser energy is output from each of a first sub-array of laser diodes. A proximal end of a fiber bundle is optically coupled to the discrete beams of laser energy. The discrete beams of laser energy are discharged through a distal end of the fiber bundle. A proximal end of a delivery fiber is optically coupled to the discrete beams of laser energy discharged through the distal end of the fiber bundle. A composite beam of laser energy comprising the discrete beams of laser energy is discharged through a distal end of the delivery fiber. In some embodiments, the shape of the composite beam is adjusted by outputting discrete beams of laser energy from a second sub-array of the laser diodes that is different from the first sub-array. In some embodiments, the method comprises adjusting a size of the composite beam by outputting discrete beams of laser energy from a second sub-array of the laser diodes that is different from the first sub-array.

This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter. The claimed subject matter is not limited to implementations that solve any or all disadvantages noted in the Background.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic diagram of a laser system in accordance with embodiments of the invention.

FIG. 2 is a simplified end view of a fiber bundle in accordance with exemplary embodiments of the invention.

FIG. 3 is a simplified end view of a fiber bundle in accordance with exemplary embodiments of the invention illustrating power intensity scalability.

FIG. 4 is a simplified cross-sectional view of an output laser beam that may be produced using the exemplary fiber bundle of FIG. 3.

FIG. 5 is a simplified end view of a fiber bundle in accordance with the exemplary embodiments of the invention that illustrates a line shaped beam.

FIG. 6 is a simplified end view of an optical fiber having a rectangular shaped core in accordance with the embodiments of the invention.

FIG. 7 illustrates a line shaped beam output from the optical fiber of FIG. 6.

FIG. 8 is a simplified end view of a fiber bundle in accordance with embodiments of the invention illustrating the production of a ring or donut shaped laser beam.

FIG. 9 is a simplified cross-sectional view of a multiple cladding layered optical fiber in accordance with embodiments of the invention.

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FIG. 10 illustrates a donut or annular shaped laser beam that may be produced using the optical fiber of FIG. 9.

FIG. 11 is a simplified end view of a fiber bundle illustrating optical fibers of the laser bundle carrying beams of laser energy produced by laser diodes having different operating modes.

FIG. 12 is a simplified cross-sectional view of an exemplary multiple cladding layered optical fiber in accordance with embodiments of the invention.

FIG. 13 is illustrated with an exemplary laser beam output from the optical fiber of FIG. 12.

FIG. 14 is a simplified end view of a fiber bundle illustrating the delivery of energy having different wavelengths in accordance with embodiments of the invention.

FIG. 15 is a simplified end view of a fiber bundle illustrating laser feedback in accordance with embodiments of the invention.

FIG. 16 is a simplified circuit diagram in accordance with embodiments of the invention.

FIG. 17 is a simplified diagram of a portion of a laser bar, in which the slow axes of the laser diodes are aligned, in accordance with embodiments of the invention.

FIG. 18 illustrates exemplary profiles of laser energy beams at surface 1 of FIG. 17.

FIG. 19 illustrates exemplary profiles of laser energy beams at surface 2 of FIG. 17.

FIG. 20 illustrates exemplary laser energy beam profiles at surface 3 of FIG. 17.

FIG. 21 is a simplified diagram of a laser bar, in which the slow axes of the laser diodes are aligned, in accordance with embodiments of the invention.

FIG. 22 illustrates laser energy beam profiles at surface 1 of FIG. 21.

FIG. 23 illustrates exemplary profiles of laser energy beams at surface 2 of FIG. 21.

FIG. 24 illustrates exemplary profiles of laser energy beams at surface 3 of FIG. 21.

FIGS. 25 and 26 show simplified side and top views, respectively, of a laser module in accordance with embodiments of the invention.

FIG. 27 illustrates an exemplary profile of the laser energy at surface 1 of FIGS. 25 and 26.

FIG. 28 illustrates an exemplary profile of the laser energy at surface 2 of FIGS. 25 and 26.

FIG. 29 is a simplified cross-sectional view of an optical fiber having a rectangular core, in accordance with embodiments of the invention.

FIG. 30 illustrates an exemplary profile of laser energy discharged from the optical fiber of FIG. 29.

FIG. 31 is an exemplary laser module in accordance with embodiments of the invention.

FIG. 32 illustrates an exemplary profile of laser energy at surface 2 of the system of FIG. 31.

FIG. 33 is a simplified diagram of a laser system in accordance with embodiments of the invention.

FIGS. 34 and 35 are simplified diagrams of laser systems in accordance with embodiments of the invention.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

Embodiments of the invention are described more fully hereinafter with reference to the accompanying drawings. Elements that are identified using the same or similar reference characters refer to the same or similar elements. The various embodiments of the invention may, however, be embodied in many different forms and should not be con-

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strued as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art.

Specific details are given in the following description to provide a thorough understanding of the embodiments. However, it is understood by those of ordinary skill in the art that the embodiments may be practiced without these specific details. For example, circuits, systems, networks, processes, frames, supports, connectors, motors, processors, and other components may not be shown, or shown in block diagram form in order to not obscure the embodiments in unnecessary detail.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

It will be understood that when an element is referred to as being “connected” or “coupled” to another element, it can be directly connected or coupled to the other element or intervening elements may be present. In contrast, if an element is referred to as being “directly connected” or “directly coupled” to another element, there are no intervening elements present.

It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. Thus, a first element could be termed a second element without departing from the teachings of the present invention.

Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

As will further be appreciated by one of skill in the art, the present invention may be embodied as methods, systems, devices, and/or computer program products, for example. Accordingly, the present invention may take the form of an entirely hardware embodiment, an entirely software embodiment or an embodiment combining software and hardware aspects. The computer program or software aspect of the present invention may comprise computer readable instructions or code stored in a computer readable medium or memory. Execution of the program instructions by one or more processors (e.g., central processing unit) results in the one or more processors performing one or more functions or method steps described herein. Any suitable patent subject matter eligible computer readable media or memory may be utilized including, for example, hard disks, CD-ROMs, optical storage devices, or magnetic storage devices. Such computer readable media or memory do not include transitory waves or signals.

The computer-usable or computer-readable medium may be, for example but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system,

apparatus, device, or propagation medium. More specific examples (a non-exhaustive list) of the computer-readable medium would include the following: an electrical connection having one or more wires, a portable computer diskette, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), an optical fiber, and a portable compact disc read-only memory (CD-ROM). Note that the computer-usable or computer-readable medium could even be paper or another suitable medium upon which the program is printed, as the program can be electronically captured, via, for example, optical scanning of the paper or other medium, then compiled, interpreted, or otherwise processed in a suitable manner, if necessary, and then stored in a computer memory.

Embodiments of the invention may also be described using flowchart illustrations and block diagrams. Although a flowchart may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be re-arranged. A process is terminated when its operations are completed, but could have additional steps not included in a figure or described herein.

It is understood that one or more of the blocks (of the flowcharts and block diagrams) may be implemented by computer program instructions. These program instructions may be provided to a processor circuit, such as a microprocessor, microcontroller or other processor, which executes the instructions to implement the functions specified in the block or blocks through a series of operational steps to be performed by the processor(s) and corresponding hardware components.

FIG. 1 is a schematic diagram of an exemplary laser system 100 in accordance with the embodiments of the invention. In some embodiments, the laser system 100 is configured to operate as a surgical laser system that generates an output beam of laser energy 102 that may be used to perform a surgical laser treatment to tissue of a patient, such as cutting, ablation, coagulation, lithotripsy or other surgical laser treatment.

In some embodiments, the system 100 includes a plurality of laser diodes 104, each of which is configured to output discrete laser energy 106. In some embodiments, the system 100 includes a fiber bundle 108 comprising a plurality of optical fibers 110, as shown in the simplified end or cross-sectional view of the fiber bundle 108 of FIG. 2. The fiber bundle 108 and the optical fibers 110 have a proximal end 112 that is coupled by way of a fiber connector 113 to the laser energy 106 output from the laser diodes 104. In some embodiments, the system 100 includes a delivery fiber 114 having a proximal end 116 that is coupled (i.e., optically coupled) to the laser energy 118 discharged through a distal end 120 of the fiber bundle 108, which comprises the laser energy 106 output from the activated laser diodes 104. In some embodiments, the composite or output laser energy 102, which comprises the laser energy 106 output from the activated laser diodes 104, is discharged through a distal end 122 of the delivery fiber 114.

In some embodiments, the system 100 includes a tubular sheath 123 having a lumen in which the delivery fiber 114 is disposed. In some embodiments, the tubular sheath is insertable into a working channel of an endoscope or cystoscope. A distal end 125 of the tubular sheath 123 is configured to facilitate the delivery of the laser energy 102 discharged from the distal end 122 of the delivery fiber 114 into a body of a patient during a surgical laser treatment.

In some embodiments, the laser energy 106 output from each of the laser diodes 104 is optically coupled to one or more of the laser fibers 110 of the fiber bundle 108 using suitable optics 124. In some embodiments, at least one of the optical fibers 110 of the fiber bundle 108 is coupled to a subset of the laser diodes 104 of the system 100 (laser diode subset or sub-array) comprising one or more laser diodes 104 using the optics 124. In some embodiments, the optics 124 include one or more optical lenses. In some embodiments, the optical lenses include a single aspheric lens and/or double lenses.

In some embodiments, the optical fibers 110 of the fiber bundle 108 may comprise different fiber subsets, each of which have different fiber properties than the optical fibers 110 of other fiber subsets. The fiber properties of the optical fibers 110 of the fiber bundle 108 may include, for example, a size of a core of the optical fiber 110, a shape of the core of the optical fiber 110, and a numerical aperture of the optical fiber 110. The exemplary fiber bundle 108 shown in FIG. 2 includes three fiber subsets: a first fiber subset comprising optical fibers 110A; a second fiber subset comprising optical fibers 110B; and a third fiber subset comprising the single optical fiber 110C. In this exemplary embodiment, the fiber subsets 110A-110C comprise optical cores of different sizes.

In some embodiments, the laser system 100 is configured to discharge laser energy 102 having different properties in order to accommodate different applications, such as different laser surgery treatments. For example, the laser system 100 may be configured to vary the wavelength, the power level or intensity, the operating mode (e.g., continuous wave or modulated/pulsed), the shape of the beam profile, and/or other properties of the output laser energy 102.

In some embodiments, optics 150 are configured to couple the proximal end 116 of the delivery fiber 114 to the laser energy 118 discharged from the distal end 120 of the fiber bundle 108, as shown in FIG. 1. In some embodiments, the optics 150 comprise one or more lenses 152.

In some embodiments, this variable output laser energy 102 is facilitated using laser diodes 104 having different laser properties. Exemplary embodiments of the laser properties include a wavelength of the laser energy 106 output by the laser diode 104, an intensity level of the laser energy 106 output by the laser diode 104, a pattern of the laser energy 106 output from the laser diode 104, a duty cycle of the laser energy output from the laser diode 104, an operating mode of the laser diode 104, and other laser properties.

In some embodiments, the system 100 includes two or more subsets or sub-arrays of the laser diodes 104 (laser diode subsets), each of which comprises one or more of the laser diodes 104 having the same or similar laser properties. In some embodiments, the laser properties of the one or more laser diodes 104 of each laser diode subset are different from the laser properties of the laser diodes 104 of other laser diode subsets. As a result, each laser diode subset is capable of producing laser energy 106 having unique properties relative to the other laser diode subsets. In some embodiments, the properties of the output laser energy 102 are adjusted through the selective activation and deactivation of one or more of the laser diode subsets.

Different applications of the output laser energy 102, such as different laser surgical treatments, often require the laser energy 102 to cover different wavelength ranges. For example, the laser energy used to ablate tissue in a benign prostatic hyperplasia (BPH) laser treatment may be different from that selected to cut tissue, ablate tissue, vaporize tissue, coagulate blood, or disintegrate kidney or bladder stones.

Green or blue laser energy having a wavelength in the range of 300-600 nm, such as 532 nm, could be useful in performing tissue ablation treatments, such as those used to treat BPH, while laser energy having a wavelength of around 2000 nm is useful in lithotripsy treatments to disintegrate kidney or bladder stones.

In some embodiments, the wavelength(s) of the composite laser energy **102** is set based on the activation of one or more laser diode subsets. For example, in some embodiments, a laser diode subset **104A** comprising one or more laser diodes **104** (labeled "LD1") are configured to output laser energy **106** having a first wavelength range (λ_1), while a laser diode subset **104B** comprises one or more laser diodes **104** (labeled "LD2") that are configured to output laser energy **106** having a second wavelength range (λ_2) that is different from the first wavelength range. Other laser diode subsets can also be used to output laser energy **106** having other unique wavelength ranges. The output laser energy **102** can be configured to include the first wavelength range through the activation of the laser diode subset **104A**, and the output laser energy **102** can be configured to include the second wavelength range through the activation of the laser diode subset **104B**. Thus, the output laser energy **102** can be configured to include one or both of the first and second wavelength ranges of laser energy **106** through the appropriate activation of one or more of the laser diode subsets **104A** and **104B**.

In one exemplary surgical application, the first laser diode subset **104A** may produce laser energy **106** having a wavelength that is strongly absorbed by hemoglobin (e.g., wavelength of 300-600 nm, such as 532 nm) and, thus, can be used to vaporize tissues containing a higher percentage of hemoglobin. The laser diode subset **104B** may produce laser energy **106** at a wavelength that is not readily absorbed by hemoglobin and can be used to coagulate tissues and stop bleeding more efficiently. Accordingly, a laser surgical treatment can be performed using the system **100** to initially vaporize targeted tissue by activating the laser diode subset **104A** to produce the output laser energy **102** that is strongly absorbed by the hemoglobin within the tissue. The system **100** can then deactivate the laser diode subset **104A** and activate the laser diode subset **104B** to produce laser energy **102** that is useful in coagulating the tissues and stopping bleeding.

The intensity or power level of the output laser energy **102** can also be adjusted through the selective activation and deactivation of one or more of the laser diode subsets. For example, when each of the laser diode subsets includes one or more laser diodes **104**, the activation of a single laser diode subset can produce the output laser energy **102** having a low power. Additional laser diode subsets can be activated to increase the intensity or power level of the output laser energy **102** resulting from an increase in the number of laser diodes **104** that are activated. As a result, the intensity or power level of the output laser energy **102** may be scaled through the activation or deactivation of the laser diode subsets. In general, the power capability of the system **100** is the sum of the power of the laser energy **106** generated by the laser diodes **104** and hence, the laser diode subsets, of the system. Accordingly, relatively low power laser diodes (e.g., 1-3 W) may be used to generate a substantially higher power laser beam **102** when collectively activated.

The laser diodes **104** or the laser diode subsets of the system **100** may also be configured to output distinct patterns of laser energy **106**. For example, one or more of laser diodes **104** may be configured to output laser energy **106** having a specific periodic pattern, such as a periodic pattern

of varying of an intensity level of the laser energy **106** (e.g., raising and/or lowering the intensity), a periodic pattern of activating and deactivating the output of the laser energy **106**, or other periodic pattern.

The laser diodes **104** or the laser diode subsets may also be configured to operate in distinct operating modes. For example, the laser diodes **104** or the laser diode subsets may be configured to operate in a continuous wave (CW) operating mode, a pulsed wave or modulated operating mode, or other conventional operating mode. In some embodiments, the laser diode subsets comprising the laser diodes **104** are configured to operate in a pulsed wave operating mode, where each laser diode subset may be configured to have unique duty cycles. The duty cycle generally operates to control the average power level of the output laser energy **106**, however, the frequency of the pulses determined by the duty cycle may also be useful in certain laser surgical treatments, such as laser lithotripsy. As a result, some embodiments of the system **100** include laser diodes **104** or laser diode subsets that operate in unique operating modes and generate laser energy **106** and output laser energy **102** having unique duty cycles.

Accordingly, it is possible to deliver laser energy **106** having different properties through different optical fibers **110** of the optical fiber bundle **108A**. FIG. 3 is a simplified end or cross-sectional view of a fiber bundle **108A** according to another embodiment of the invention, in which each of the optical fibers **110** has been labeled with a number 1-5 to designate a laser diode subset or sub-array to which they are coupled. That is, the optical fibers **110** that are numbered "4" each are coupled to the laser energy **106** discharged from the laser diodes **104** of one laser diode subset or sub-array, while the optical fibers numbered "5" are each coupled to the laser energy **106** discharged from the laser diodes **104** of another laser diode subset or sub-array. Accordingly, the activation of one or more of the subsets of laser diodes **104** delivers the corresponding laser energy **106** through the corresponding optical fibers **110** and through the delivery fiber **114** as the output laser energy **102**.

As a result, the properties of the output laser energy **102** may be customized or tuned through the activation and deactivation of the laser diodes **104** or the laser diode subsets. For example, the system **100** may be operated to activate the laser diode subsets, as indicated by the shaded optical optical fibers **110** (optical fibers 1, 2, 3 and 5), while the laser fiber subset corresponding to the optical fibers 4 is deactivated. This results in composite or output laser energy **102** discharged from the delivery fiber **114** that comprises the laser energy **106** generated by the laser diode subsets corresponding to the optical fibers 1-3 and 5.

In some embodiments, the activation and deactivation of different laser diode subsets controls the size of the beam of output laser energy **102** discharged from the laser fiber **114**. For example, the laser diode subsets corresponding to the optical fibers 1 and 2 (FIG. 3) may produce an output laser beam **102** having a relatively small diameter, which can be increased by activating other laser diode subsets, such as the laser diode subset corresponding to optical fibers 3-5, as shown in FIG. 4. In some embodiments, when the laser energy **106** output from the laser diodes **104** has substantially the same intensity level, the increase in the size of the diameter of the discharge laser energy **102** maintains a substantially even distribution of the laser energy. The larger diameter beam can be used to remove tissue more quickly while the smaller sized beam can be used to remove tissue more precisely.

In some embodiments, the shape of the output laser beam **102** discharged from the delivery fiber **114** may be chosen or adjusted through the activation of select laser diode subsets and/or the configuration of the delivery fiber **114**. In some embodiments, the delivery fiber **114** comprises an optical fiber having a round core, such as a conventional optical fiber, which discharges the laser energy **102** in a circular shaped beam **102**.

FIG. **5** is a simplified end or cross-sectional view of a fiber bundle **108B** in accordance with exemplary embodiments of the invention. As illustrated in FIG. **5**, the activation of laser diode subsets **1** and **2** deliver laser energy **106** through optical fibers **110** that are oriented in a line. In some embodiments, the delivery fiber **114A** comprises an optical fiber having a rectangular core **130** surrounded by cladding **132**, as shown in the simplified end or cross-sectional view of FIG. **6**. In some embodiments, the index of refraction (n_1) of core **130** is greater than the index of refraction (n_2) of the cladding **132**. The rectangular shaped core **130** allows the delivery fiber **114A** to deliver a line shaped output beam **102** to a desired target. FIG. **7** illustrates an exemplary line shaped output beam **102** that may be discharged from the optical fiber of FIG. **6** as simulated using ZEMAX (optical simulation software). In some embodiments, the line shaped laser beam **102** can be used in a surgical laser procedure to enucleate tissues. When the line shaped output beam **102** is swept across tissue, it can also be used to vaporize the tissue more precisely than round shaped laser beams.

In some embodiments, the system **100** is configured to discharge an annular or donut shaped output beam **102**. In some embodiments, this is accomplished by activating laser diode subsets corresponding to optical fibers **110** of the fiber bundle **108C** that form an annular or ring pattern. For example, the activation of the laser diode subsets **4** and **5** that deliver laser energy **106** to the corresponding optical fibers **110** of the fiber bundle **108C** (FIG. **8**) results in a delivery of an annular or donut shaped beam of laser energy **118** to the delivery fiber **114**. In some embodiments, the delivery fiber **114** is configured to discharge this annular laser energy as an annular, ring- or donut-shaped beam **102**.

In some embodiments, the delivery fiber **114B** may comprise a multiple cladding optical fiber, such as that shown in the simplified cross-sectional view of FIG. **9**. In some embodiments, the multiple cladding optical fiber **133** comprises central cladding **134** having an index of refraction (n_1) that is lower than the index of refraction (n_2) of an annular light delivery medium **136**. Additionally, the index of refraction (n_2) of the annular light delivery medium **136** is greater than the index of refraction of outer cladding **138**. In some embodiments, the annularly shaped energy **106** discharged from the fiber bundle **108C** is optically coupled to the annular light delivery medium **136** of the delivery fiber **114B** and is discharged through the distal end **122** as an annularly or donut-shaped output beam **102**, a simulation of which is illustrated in FIG. **10** using the ZEMAX software. In some embodiments, the annularly shaped output beam **102** is used in a surgical laser procedure to enucleate tissues.

FIG. **11** is a simplified end or cross-sectional view of a fiber bundle **108D** delivering laser energy in accordance with embodiments of the invention. In some embodiments, the laser diode subsets **1**, **2**, **4** and **5** are activated while the laser diode subset **3** is deactivated. Thus, laser energy **106** generated by the subsets **1**, **2**, **4** and **5** is delivered through the corresponding optical fibers **110** of the fiber bundle **108D**.

In some embodiments, the delivery fiber **114C** is a form of a multiple clad fiber **140**, a simplified cross-sectional view

of which is provided in FIG. **12**. In some embodiments, the multiple cladding optical fiber **140** comprises a central light delivery medium **142**, a cladding **144**, an annular light delivery medium **146** and a cladding **148**, as shown in FIG. **12**. In some embodiments, the central light delivery medium **142** and the annular light delivery medium **146** comprise glass. In some embodiments, the cladding **144** surrounds the central light delivery medium **142** and has an index of refraction (n_2) that is less than the index of refraction (n_1) of the central light delivery medium **142**. The annular light delivery medium **146** surrounds the cladding **144**, and the cladding **148** surrounds the annular light delivery medium **146**. In some embodiments, the index of refraction (n_2) of the cladding **144** and the cladding **148** are less than an index of refraction of the annular light delivery medium **146**.

In some embodiments, in order to deliver the laser energy from the embodiment of the fiber bundle **108D** depicted in FIG. **11**, the delivery fiber **114C** depicted in FIG. **12** is used. In this embodiment, the laser energy **106** delivered by the optical fibers **110** of the fiber bundle **108D** corresponding to the laser diode subsets **1** and **2** (FIG. **11**) is coupled to the central light delivery medium **142**, and the laser energy **106** delivered by the optical fibers **110** of the fiber bundle **108D** corresponding to the laser diode subsets **4** and **5** is coupled to the annular light delivery medium **146**. This configuration allows the delivery fiber **114C** to deliver the laser energy **106** in the form of a beam **102** having a central circular portion and an annular portion as illustrated in FIG. **13**, which is a simulation produced using the ZEMAX application.

As mentioned above, the laser diodes **104** or the laser diode subsets may be operated to produce laser energy **106** having different properties. For example, one or more laser diode subset may operate in a continuous wave mode or a high duty cycle to produce high intensity or high average power laser energy **106**, while other laser diodes **104** or laser diode subsets may be modulated at a certain frequency or duty cycle to produce laser energy **106** having a lower average power or intensity. In some embodiments, the laser diode subsets **1** and **2** (FIG. **11**) may be operated in a high powered mode (continuous wave or high duty cycle), while the laser diode subsets **4** and **5** may be modulated to produce a relatively low average power laser energy **106**. As a result, the high powered laser energy is centrally located in the fiber bundle **108D** while the low powered laser energy is located at the periphery of the fiber bundle **108D**. When the delivery fiber **114C** of FIG. **12** is used, the high power laser energy **106** is coupled to the central light delivery medium **142**, while the lower power laser energy **106** is coupled to the annular light delivery medium **146**. The resultant output beam **102** may be used in a surgical procedure in which the central beam cuts or vaporizes tissues while the outer annular beam simultaneously coagulates tissues.

In some embodiments, the system **100** is configured to deliver laser energy **106** generated by one or more subsets of the laser diodes **104** having a wavelength configured to vaporize tissue, while an inner cluster of the optical fibers **110** are configured to deliver laser energy **106** from one or more subsets of the laser diodes **104** having a wavelength that is configured to coagulate tissue, as shown in FIG. **14**, which is a simplified end or cross-sectional view of an exemplary fiber bundle **108E** in accordance with embodiments of the invention.

In some embodiments, the system **100** is configured to provide electromagnetic energy feedback for identification, diagnosis, or other purposes. In some embodiments, the one or more laser diodes **104** include an excitation laser diode **104E** (FIG. **1**) that is configured to output excitation laser

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energy **106** having a wavelength in an excitation spectrum, as shown in FIG. **1** and FIG. **14**, which is a simplified end or cross-sectional view of the fiber bundle **108E** in accordance with embodiments of the invention. The excitation laser energy **106** generated by the laser diode **104E** is delivered to a target through at least one of the optical fibers **110** of the fiber bundle **108F**, such as laser fiber **110E** shown in FIG. **15**, and the delivery fiber **114**. In some embodiments, the excitation laser energy is combined with laser energy generated from one or more other subsets of the laser diodes **104** and is output as the laser energy **102** from the delivery fiber **114**.

In some embodiments, the laser energy generated by the excitation laser diode **104E** is transmitted through a band-pass filter **154** (FIG. **1**) to ensure that the excitation laser energy is within the desired wavelength range of the excitation spectrum. In some embodiments, the excitation spectrum is in the range of 300-420 nanometers.

In some embodiments, the excitation laser energy is configured to target tissue or other substance that generates auto-fluorescence electromagnetic energy or feedback electromagnetic energy **158** in response to the exposure to the excitation laser energy that can be used to identify the tissue or substance, diagnose a condition of the tissue or substance, or used for other purposes. The feedback electromagnetic energy **158** is captured by the delivery fiber **114** at the distal end **122** and transmitted through the delivery fiber **114** and at least one of the optical fibers **110** of the fiber bundle **108F**, such as optical fiber **110F** shown in FIG. **15**. Co-pending, commonly assigned International Application No. PCT/US14/61319, filed Oct. 20, 2014, the contents of which are incorporated by reference herein in its entirety for all purposes, discloses additional devices and methods to identify conditions at the treatment site.

In some embodiments, as depicted in FIG. **2**, the system **100** includes a spectrometer **156** that is configured to analyze the feedback electromagnetic energy **158**. In some embodiments, the system **100** includes a filter **160** that is configured to filter the feedback electromagnetic energy **158**, and the filtered feedback electromagnetic energy **158** which is then analyzed by the spectrometer **156**. In some embodiments, the filter **160** is configured to remove the excitation spectrum and/or isolate a desired auto-fluorescence spectrum of the feedback electromagnetic energy **158**.

FIG. **16** is a schematic diagram of an exemplary circuit used to facilitate one or more embodiments of the system **100** described above. In some embodiments, laser diodes **104** are arranged for individual or group activation by coupling a current from a power supply **170**, which receives power from a suitable power source **171**, to the individual or group of laser diodes **104** by shorting out individual or groups (subsets) of laser diodes using appropriate switches **172** (e.g., MOSFETs) responsive to control signals from a controller **174** comprising one or more processors. In some embodiments, control circuits **173** for the MOSFETs or switches **172**, such as opto isolators, may be used to process the control signals from the controller **174**.

In some embodiments, monitoring circuitry **176** is provided to monitor the performance of each of the laser diodes **104** either optically or by measuring a voltage drop across the diodes **104**. This enables defective laser diodes **104** to be detected and isolated. In some embodiments, the circuitry **176** measures a voltage drop across the individual laser diodes **104** or a group of laser diodes **104** as feedback for the performance of the laser diodes **104**.

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In some embodiments, the circuitry includes a switch **178** (e.g., a MOSFET) that is configured to enable or disable current through the set of laser diodes **104**.

In some embodiments, the system **100** includes control loop circuitry **180** for controlling the power supply **170**. In some embodiments, the control loop circuitry **180** receives a voltage drop across a current sense resistor **182**, such as a 4-terminal or a 2-terminal resistor, which is used to determine the current through the activated laser diodes **104**. In some embodiments, the control loop circuitry **180** includes an optical power feedback, which indicates the intensity or power of the laser energy **102** delivered to a target. In some embodiments, the control loop circuitry **180** includes a set point from the controller **174** which indicates a desired power setting.

Some embodiments of the invention are directed to a method of producing a laser beam **102** using a laser system **100** in accordance with one or more embodiments described herein. In some embodiments of the method, a discreet beam of laser energy **106** is output from each of a plurality of laser diodes **104** of the system **100**. A proximal end **112** of a fiber bundle **108** is optically coupled to the discreet beams of laser energy **106**. The discreet beams of laser energy **106** are discharged through a distal end **120** of the fiber bundle **108** (laser energy **118**). A proximal end **116** of a delivery fiber **114** is optically coupled to the discreet beams of laser energy **106** discharged through the distal end **120** of the fiber bundle **108**. A composite beam of laser energy **102** comprising the discreet beams of laser energy **106** is discharged through a distal end **122** of the delivery fiber **114**, which is optically coupled to the fiber bundle **108**.

In some embodiments, outputting or generating the discreet beams of laser energy **106** from each of a plurality of the laser diodes **104** of the system **100** comprises outputting a beam of laser energy **106** from a first subset of the laser diodes **104**.

In some embodiments, the intensity of the composite beam **102** is adjusted by outputting or generating discreet beams of laser energy **106** from a second subset of the laser diodes **104** that is different from the first subset of the laser diodes **104**. In accordance with this step of the method, the total number of laser diodes **104** that are activated by the system **100** to generate the discreet beams of laser energy **106** may be either increased or decreased to adjust the total power level of the composite beam of laser energy **102** discharged from the delivery fiber **114**. This embodiment allows the surgeon to transition the laser energy **102** between high and low intensity modes, such as from a coagulation mode to a vaporization or cutting mode, or from an aiming beam mode, in which the laser energy **102** lacks sufficient intensity to damage tissue of the patient, to an active mode, in which the laser energy **102** has sufficient intensity to cut, vaporize or perform another laser procedure on tissue of the patient.

In some embodiments of the method, a wavelength or wavelengths of the composite beam of laser energy **102** is adjusted by outputting or generating discreet beams of energy **106** from a second set of laser diodes that is different from the first subset. In this embodiment, the first laser diode subset may produce a composite beam of laser energy **102** spanning a first set of wavelengths, whereas activation of the second laser diode subset results in a composite laser beam **102** spanning a different set of wavelengths. This embodiment allows the surgeon to transition the laser energy **102** between wavelengths that are useful for one type of laser procedure, to wavelengths that are useful in performing another type of laser procedure, for example. For instance,

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some wavelengths of the laser energy **102** are useful in vaporizing tissue (e.g., 532 nm), while other wavelengths of the laser energy **102** may be more useful in ablating or cutting tissue.

In some embodiments of the method, a size of the composite laser beam **102** is adjusted by outputting discrete beams of laser energy **106** from the second subset of the laser diodes **104** that is different from the first laser diode subset. As discussed above with reference to FIGS. **3** and **4**, this allows the laser system **100** to adjust to the diameter of the composite laser beam **102**. Additional techniques for adjusting the size and shape of the composite laser beam **102** discharged from the delivery fiber **114** involve selecting a delivery fiber **114** having an optical fiber that results in the desired size and shape of the discharge composite laser beam **102**.

In some embodiments of the method, a shape of the composite laser beam **102** is adjusted by outputting discrete beams of laser energy **106** from a second subset of the laser diodes **104** that is different from the first laser diode subset. For example, a composite beam **102** having a shape of a circle (FIGS. **3** and **4**), a ring (FIGS. **8-10**), a line (FIGS. **5-7**), a square, a rectangle, and concentric rings or circles (FIGS. **11-13**), or other desired shape for the composite laser beam **102** may be realized through the selected activation and deactivation of subsets of the laser diodes **104**.

In some embodiments of the method, a pattern of the composite laser beam **102** can be adjusted by outputting discrete beams of laser energy **106** from a second subset of the laser diodes that is different from the first laser diode subset. As discussed above, this may involve a periodic variation to the laser energy **106** discharged from the laser diodes **104** of the first laser diode subset as compared to the laser diodes **104** of the second laser diode subset.

In some embodiments of the method, the proximal end **112** of the fiber bundle **108** is optically coupled to the discrete beams of laser energy **106** by optically coupling the proximal end **112** of the fiber bundle **108** to the discrete beams of laser energy **106** using optics **124** comprising one or more lenses. In some embodiments of the method, the discrete beams of laser energy **106** are discharged through the distal end **120** of one or more of the optical fibers **110** of the fiber bundle **108**.

In some embodiments of the method, optically coupling a proximal end **116** of the delivery fiber **114** to the discrete beams of laser energy **106** (laser energy **118**) discharged through the distal end **120** of the fiber bundle **108** comprises optically coupling proximal end **116** of the delivery fiber **114** to the discrete beams of laser energy **106** discharged through the distal end **120** of the fiber bundle **108** using optics **150** comprising one or more lenses **152**.

In some embodiments of the method, outputting a discrete beam of laser energy **106** from each of a plurality of the laser diodes **104** of the system **100** comprises outputting one or more discrete beams of excitation laser energy **106** having a wavelength within an excitation spectrum. In some embodiments, the method includes outputting one or more discrete beams of laser energy **106** having a wavelength within the excitation spectrum. In some embodiments, one or more of the discrete beams of laser energy **106** are passed through a filter **154** that filters one or more of the discrete beams of laser energy **106** such that they are within the excitation spectrum. In some embodiments, the excitation spectrum is 300-420 nanometers.

In some embodiments of the method, feedback electromagnetic energy **158** is transmitted from the distal end **122** of the delivery fiber **114** through the delivery fiber **114** and

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the fiber bundle **108** responsive to the exposure of tissue or a substance to the excitation laser energy. In some embodiments, the feedback electromagnetic energy **158** is delivered to a spectrometer **156** (FIG. **1**). In some embodiments, the feedback electromagnetic energy **158** is filtered using a filter **160** before the feedback electromagnetic energy **158** is delivered to the spectrometer **156**.

FIGS. **17** and **21** are simplified diagrams of laser bars **200** in accordance with embodiments of the invention. The laser bars **200** are configured to discharge a composite beam of laser energy **202** that is formed from laser energy discharged from a plurality of laser diodes **204**. In some embodiments, the laser bar **200** includes a plurality of collimating lenses **208**, each of which is configured to collimate the laser energy **206** output from one of the laser diodes **204** into a collimated beam of laser energy **210**. In some embodiments, the collimating lenses **208** are aspheric lenses having a larger numerical aperture ($NA > 0.6$).

In some embodiments, the laser bar **200** includes at least one minor **212** configured to reflect the collimated beams **210**. In some embodiments, the laser bar **200** includes optics **214** configured to gather the reflected collimated beams **210** into the composite beam **202**, which may be discharged from the laser bar **200** to an optical fiber for delivery to a target. In some embodiments, the at least one mirror **212** comprises an individual minor for each laser diode **204** that is configured to reflect the corresponding collimated beam **210** toward the optics **214**, as shown in FIGS. **17** and **21**. In some embodiments, each of the minors **212** are angled at approximately 45 degrees to the corresponding collimated beam **210**.

In some embodiments, the spacing between each laser diode **204** in the X direction (as identified in the figures) is made as small as possible and is generally determined by the package of the laser diode **204**. In some embodiments, the package for each laser diode **204** has a diameter of approximately 5.6 millimeters and the spacing along the X axis between individual laser diodes **204** is approximately 6 millimeters.

In some embodiments, the laser diodes **204** and their corresponding minors **212** are positioned such that the laser energy **206** discharged from each of the laser diodes **204** travels approximately the same total distance from the laser diodes **204** to the optics **214**. In some embodiments, the minors **212** are displaced from the laser diodes **204** along the Z axis, and the mirrors **212** are displaced from the optics **214** along the X axis that is perpendicular to the Z axis, as shown in FIGS. **17** and **21**.

In some embodiments, the laser energy output from the laser diodes **204** is asymmetrical. In some embodiments, the beam shape of the laser energy **206** discharged from each laser diode **204** is elliptical, as the fast axis along the pn junction has a much larger divergent angle compared to the slow axis, which is perpendicular to the pn junction.

In some embodiments, the fast axes of the laser diodes are aligned in the X-Z plane, as shown in FIG. **17**, and the laser energy **206** output from each of the laser diodes **204** and the corresponding collimated beams **210** are substantially aligned in the X-Z plane. This results in an elliptical beam profile at surface **215**, as shown in FIG. **18**, which is a simulation of the beam profiles produced using the ZEMAX software. The beam profiles at surface **217** of FIG. **17** are illustrated in FIG. **19**.

In some embodiments, the slow axes of the laser diodes **204** are aligned in the X-Z plane, as shown in FIG. **21**, and the laser energy **206** and the corresponding collimated beams **210** are substantially aligned in the X-Z plane. This

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results in the beam profiles shown in FIG. 22 at surface 215, and the beam profiles shown in FIG. 23 at surface 217.

In some embodiments, the light combining optics 214 operate to reduce asymmetry of the collimated beams 210 reflected from the minors 212. In some embodiments, the light combining optics 214 include cylindrical lenses 216 and 218, which operate to reduce the fast axis beam divergent angle of each beam 210 and make it the same as the slow axis beam divergent angle. As a result, the individual beams output from the beam gathering optics 214 at surface 219 have substantially circular profiles, as shown in FIG. 20 (beam profiles at surface 219 of FIG. 17) and FIG. 24 (beam profiles at surface 219 of FIG. 21).

Some embodiments of the invention are directed to a laser module comprising two or more of the laser bars 200 formed in accordance with one or more of the embodiments described herein. FIGS. 25 and 26 show simplified side and top views, respectively, of an exemplary laser module 300 in accordance with embodiments of the invention, which includes eight laser bars 200. However, embodiments of the invention include laser modules 300 having more or fewer laser bars 200.

In some embodiments, the laser module 300 includes at least one minor 302 that is configured to reflect the composite beams 202 of the laser bars 200. An exemplary profile of the reflected composite beams 202 of each beam bar 200 at surface 303 of FIGS. 25 and 26 is shown in FIG. 27. In some embodiments, the at least one minor 302 comprises a plurality of mirrors 302 that are each configured to reflect one of the composite beams 202, as shown in FIG. 26. In some embodiments, each of the minors 302 is angled at approximately 45 degrees to their corresponding composite beam 202. In some embodiments, the optical distance of each laser bar 200 and the corresponding minor 302 is optimized so that the optical distance from each laser bar 200 to surface 303 is the same.

In some embodiments, each of the mirrors 302 is displaced relative to the other mirrors 302 in the X direction. Preferably, this height difference of each of the minors 302 is as small as possible and is generally determined by the diameter of the bars 200. In some embodiments, each of the laser bars 200 has a diameter of approximately 1.2 millimeters, and the mirrors 302 are separated from each other in the X direction by approximately 1.4 millimeters.

In some embodiments, the laser module 300 includes light focusing optics 304 configured to focus the reflected composite beams 202 into a convergent beam 306. In some embodiments, the light focusing optics 304 comprises a single spherical lens, as indicated in FIGS. 25 and 26. The resultant profile of the convergent beam 306 at surface 307 is shown in FIG. 28. The profile of the convergent beam 306 is elliptical after the spherical focusing lens 304 due to the wider dispersion angle of the laser energy along the fast axis of the beam 102 as compared to that along the slow axis.

In some embodiments, the laser module 300 includes an optical fiber 308 having a proximal end 310 optically coupled to the convergent beam 306. The optical fiber 308 may operate similarly to the delivery fibers 114 described above, and includes a distal end 312 through which the convergent beam 306 is discharged. The distal ends of the delivery fibers disclosed herein may have an end-firing configuration or a side-firing configuration (i.e., a beveled distal end), depending of the laser treatment/procedure being performed.

In some embodiments, the optical fiber 308 has a rectangular core 320 surrounded by cladding 322, as shown in the simplified cross-sectional view of FIG. 29. Such an optical

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fiber 308 can be used to deliver the elliptically shaped convergent beam 306 at the proximal end 310 of the fiber 308, which can be discharged from the delivery end of the optical fiber 308. In some embodiments, the rectangular core 320 has dimensions of 750 μm by 200 μm . An exemplary profile of the beam discharged from the optical fiber 308 having the rectangular core 320 is shown in FIG. 30. Such a rectangular or line shaped beam can behave like an optical knife for tissue cutting. Furthermore, by sweeping the line shaped beam, it can be used to vaporize tissues.

In some embodiments, as depicted in FIG. 31, the light focusing optics 304 comprise two cylindrical lenses 304A and 304B to increase the launching beam area in front of the receiving fiber 308, and to make the beam 306 substantially square or rectangular in shape. An exemplary profile of the beam 306 at surface 307 of FIG. 31 is shown in FIG. 32. In some embodiments, a circular core shaped optical fiber 308 can be used to receive the beam 306 at the proximal end 310 and discharge a circular beam at the output end of the optical fiber 308.

Some embodiments of the invention are directed to a laser system 400, in which two or more of the laser modules 300 are combined. FIG. 33 is a simplified diagram of such a laser system 400 in accordance with embodiments of the invention. In some embodiments, the laser system 400 includes a polarization beam combining cube 402 that receives each of the composite output beams 306 from the modules 300 and outputs a composite beam 404 comprising each of the beams 306, as shown in FIG. 33. In some embodiments, one or more of the laser modules 300 are configured to output their beams 306 in the Y-direction, and one or more of the laser modules 300 are configured to output their beams 306 in the X-direction, as shown in FIG. 33.

In some embodiments, the X-direction beams 306 and the Y-direction beams 306 are linear-polarized beams with 90 degree polarization direction difference there-between. This can be implemented by having one set of the modules 300 using fast axis beam aligned bars 200 while the other group of modules 300 use slow axis beam aligned bars 200.

In some embodiments, the laser system 400 includes focusing optics 406, which are configured to couple the composite beam 404 to a proximal end 408 of an optical fiber 410, which can then discharge the beam 404 to a desired target.

The free space optical beam combining achieved by the laser system 400 can be used to make a modularized device. Simplified diagrams of examples of such laser systems are illustrated in FIGS. 34 and 35. For example, as shown in FIG. 34, a beam array (M×N) can be formed using M sub-modules (each sub-module has an N×1 beam bar 200). To manufacture this system 400, a sub-module with N×1 beam bars 200 (where N is 2, 3, 4 . . . n) is made based on a customer's requirement, M sub-modules (where M is 1, 2, 3 . . . m) is plugged into the laser console to obtain the required output power (M×N times of single diode power). The power intensity is dependent on the output power and the delivery fiber core shape and size.

As shown in FIG. 35, another option is to make an N×N sub-module of the laser bars 200 and plug M of this module into a laser console based on output power requirements. The overall power is M×N×N times of a single laser diode power.

As mentioned above, the above-described laser systems and devices are useful generating and delivering laser energy that may be used to perform a surgical laser treatment on a patient. Exemplary surgical laser treatments include cutting, ablation, coagulation, lithotripsy, or other treatment.

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Embodiments of the invention include the performance of such surgical laser treatments on a patient using embodiments of the systems or devices described herein.

Although the present invention has been described with reference to preferred embodiments, workers skilled in the art will recognize that changes may be made in form and detail without departing from the spirit and scope of the invention.

What is claimed is:

1. A controller for a surgical laser system, the surgical laser system including an array of laser diodes optically couplable to a delivery fiber, wherein the controller is configured to:

activate a first sub-array of the array of laser diodes to cause each laser diode of the first sub-array of laser diodes to output a first laser energy at a first wavelength range;

activate a second sub-array of the array of laser diodes to cause each laser diode of the second sub-array of laser diodes to output a second laser energy at a second wavelength range;

receive, via optical monitoring circuitry coupled to the array of laser diodes, an indication of an optical power of an output laser beam delivered to a target via the delivery fiber, wherein the output laser beam is a composite laser beam comprising the first laser energy output by the laser diodes of the first sub-array of laser diodes and the second laser energy output by the laser diodes of the second sub-array of laser diodes; and

activate, responsive to the optical power, a third sub-array of the array of laser diodes to cause each laser diode of the third sub-array to output a third laser energy at the first wavelength range or the second wavelength range; or

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deactivate, responsive to the optical power, the first sub-array of laser diodes and/or the second sub-array of laser diodes.

2. The controller of claim 1, wherein the controller is configured to control the first sub-array of laser diodes and the second sub-array of laser diodes to operate in different operating modes, wherein the different operating modes include one or more of a continuous wave operating mode, a pulsed wave operating mode, or a modulated operating mode.

3. The controller of claim 1, wherein the controller is further configured to selectively activate or deactivate additional sub-arrays of laser diodes to increase or decrease a power level of the output laser beam.

4. The controller of claim 1, wherein the controller is configured to control the first sub-array of laser diodes and the second sub-array of laser diodes to output laser energies with different periodic patterns.

5. The controller of claim 1, wherein the first sub-array of laser diodes and the second sub-array of laser diodes are configured such that activation of the first sub-array results in a discharge of a first beam of laser energy, and simultaneous activation of the first and second sub-arrays results in a discharge of a second beam of laser energy having a different size or shape than the first beam.

6. The controller of claim 1, wherein the controller is a user-actuated controller configured for selective activation and deactivation of the first sub-array of laser diodes and the second sub-array of laser diodes based on input received from a user.

7. The controller of claim 1, wherein the controller is configured to activate the first sub-array of laser diodes at a first selected power setting and/or to activate the second sub-array of laser diodes at a second selected power setting.

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